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# Classification of hyperbolic space-like constant mean curvature surfaces in Minkowski space

**Tesi di Laurea Magistrale in Matematica**

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# Abstract

The thesis is aimed at the study of complete hyperbolic space-like constant mean curvature surfaces (CMCs) in Minkowski 3-space. After introducing the necessary preliminary material on space-like surfaces in Minkowski space, harmonic maps, and holomorphic quadratic differentials, our work focuses on translating the geometric properties derived from the constant mean curvature hypothesis into analytic properties.

In particular, we construct a map that associates a holomorphic quadratic differential on the Poincaré disk to each CMC cut. We then invert this correspondence, proving that this map is a bijection using the previously established properties. This reduces the problem to the study of a nonlinear PDE for the conformal factor of the induced metric. The last part of the thesis is therefore devoted to proving existence, by means of a sub- and supersolution method, and uniqueness, by means of a generalized maximum principle.

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# Chapter 1

# Preface

The first part of the thesis is focused on the geometric background. We begin by considering space-like immersions in Minkowski space and discussing some global properties of complete immersions. In particular, the hypothesis of completeness implies that these surfaces are simply connected. After endowing these surfaces with a natural complex structure, we can then use the uniformization theorem to parametrize them by their universal cover, which is a concrete object, namely the Poincaré disk, on which all the relevant objects can be studied.

We then give a different formulation of the Gauss–Codazzi equations, adapting them to the Lorentzian setting. This yields a form of the equations that differs in a small yet important way from the classical Euclidean case. Furthermore, these equations play an important role in the later development of the thesis.

Another crucial step is to prove that the Gauss map of a space-like surface, whose values in this case lie in $\mathbb{H}$ instead of the classical $\mathbb{S}^2$, is harmonic if and only if the mean curvature is constant. This fact is important because it translates a geometric condition on the surface into an analytic condition on its Gauss map. In turn, this implies that the corresponding Hopf differential is holomorphic, and this leads to the construction of a map

$$\mathscr{A} : \mathbf{E} \to \mathbf{Q}_{\mathbb{D}},$$

which associates to each CMC-1 cut, up to isometry, a holomorphic quadratic differential on the Poincaré disk.

The main result of this work is to show that this map is a bijection. In order to prove it, we have to invert the previous construction. Hence, starting from a holomorphic quadratic differential, we reconstruct the geometric data of a surface, namely the first and second fundamental forms. This is performed by using the properties found in the previous step, which are necessary conditions and turn out also to be sufficient. First, the Gauss equation, which we translated into a PDE, is solved for the conformal factor of the induced metric, obtaining the first fundamental form. Then, using the properties of traceless symmetric tensors, we reconstruct the second fundamental form.To reconstruct the surface itself, we use Bonnet's fundamental theorem, which requires that the

candidate first and second fundamental forms satisfy the Gauss-Codazzi equations. Gauss' equation is satisfied by construction, since the conformal factor is obtained by solving the corresponding PDE. The Codazzi equations are reduced to showing that a traceless symmetric 2-tensor satisfies the Codazzi equation if and only if the associated quadratic differential is holomorphic, which is true by assumption.

The last part of the thesis is dedicated to the study of the aforementioned PDE. The bijectivity of $\mathscr{A}$ is reduced to existence and uniqueness for this equation under the necessary conditions of completeness of the induced metric of $S$ and positivity of the Jacobian of the respective Gauss map. Uniqueness is obtained by means of a generalized maximum principle, while existence is proved through a sub- and supersolution method, first on bounded domains and then by exhaustion of the domain with bounded sets.

Our discussion is based mainly on the work of Wan [10].

# Chapter 2

# Minkowski space and space-like surfaces

The environment we will work within is the Minkowki 3-space, defined as $\mathbb{R}^3$ endowed with the pseudometric

$$ds^2 = (dx^1)^2 + (dx^2)^2 - (dx^3)^2.$$

In order to distinguish the use of this metric from the standard euclidean metric of $\mathbb{R}^3$ we will use the notation $\mathbb{M}^{2,1}$. We will also denote the coordinates either as $(x^1, x^2, x^3)$ or $(x, y, t)$, choosing the notation according to convenience and will identify the slice $\{(x, y, 0)\}$ of $\mathbb{M}^{2,1}$ with $\mathbb{R}^2$ and with $\mathbb{C}$.

The space $\mathbb{M}^{2,1}$ is an example of a more general object called a pseudo-Euclidean vector space, a term that refers to a finite-dimensional vector space $V$ endowed with a nondegenerate (possibly not positive-definite) symmetric bilinear form $g$. This kind of space serves as the "infinitesimal model" for pseudo-Riemannian manifolds, smooth manifolds endowed with a smooth non-degenerate 0-2 tensor. If the signature of this tensor is exactly $(n-1, 1)$, the manifold is called a Lorentzian manifold.
The following definition, borrowed from physics, will be useful later.

*Definition.* Given $(M, g)$ a Lorentzian manifold and $p \in M$, we say that a vector $v \in T_pM$ is

- space-like if $\langle v, v\rangle > 0$,
- light-like if $\langle v, v\rangle = 0$ and $v \neq 0$,
- time-like if $\langle v, v\rangle < 0$.

We also call a subspace $P \subseteq T_pM$ space-like if every nonzero vector in $P$ is space-like.

## 2.1 Space-like immersions

We will always denote with $S$ a differentiable manifold of dimension 2 (a surface) immersed in $\mathbb{M}^{2,1}$, $\iota : S \hookrightarrow \mathbb{M}^{2,1}$. We assume that the ambient pseudometric,

$g_{\mathbb{M}^{2,1}}$ when restricted to $S$, yields a nondegenerate positive definite bilinear form. We call this form the first fundamental form or pull-back pseudometric, which we indicate as $\mathbf{I} = \iota^*(g_{\mathbb{M}^{2,1}})$ and write in coordinates as $\mathbf{I} = g_{ij}dx^i dx^j$ (for simplicity we sometimes forget the immersion $\iota$). By definition, this condition is equivalent to requiring that every tangent plane to $S$ be space-like, and such a surface is called a space-like surface.

A nice property of these surfaces is that they are locally the graph of a function of the form $x^3 = f(x^1, x^2)$, $f$ being smooth and Lipschitz with constant less than 1.
We can show this with a simple argument:
First, we assumed that the restriction of the Lorentzian metric to each tangent space $T_pS$ is positive-definite so that it is a space-like plane. This can be rewritten as

$$T_pS \subset \mathscr{C} := \{v \in \mathbb{M}^{2,1} : \langle v, v\rangle \geq 0\} = \{t^2 \leq x^2 + y^2\} \ .$$

The set $\mathscr{C}$ is the outside-region of the double cone in $\mathbb{M}^{2,1}$ whose axis is $\{x^1 = 0,\ x^2 = 0\}$. Clearly $\mathscr{C}$ does not contain any vertical vector and clearly the same holds for $T_pM$. Therefore we can apply the implicit function theorem and express $S$ locally near $p$ as the graph of a function $t = f(x, y)$, where $f : U \subset \mathbb{R}^2 \to \mathbb{R}$. Where this local representation is valid, we can parametrize locally the surface as $(x, y, f(x, y))$, and obtain the local frame

$$V_x = (1, 0, \partial_x f(x, y)), \qquad V_y = (0, 1, \partial_y f(x, y)),$$

for which the first fundamental form is represented by the matrix

$$g = \begin{pmatrix} \langle V_x, V_x\rangle & \langle V_x, V_y\rangle \\ \langle V_y, V_x\rangle & \langle V_y, V_y\rangle \end{pmatrix} = \begin{pmatrix} 1 - f_x^2 & -f_x f_y \\ -f_x f_y & 1 - f_y^2 \end{pmatrix}.$$

Since the parametrization is regular, the condition that $g$ be positive-definite is equivalent to

$$0 < \det g = 1 - (f_x^2 + f_y^2) = 1 - |\nabla f|^2,$$

and therefore $|\nabla f|^2 < 1$.

## 2.2 Entire surfaces

Given a surface $S$ immersed in $\mathbb{M}^{2,1}$, we can define a projection $\Pi : S \to \mathbb{R}^2$ by forgetting the time coordinate, $\Pi(x^1, x^2, x^3) := (x^1, x^2)$ for any $(x^1, x^2, x^3) \in S$. We call the surface $S$ is entire when the map $\Pi$ is a diffeomorphism. If the surface is globally a graph of a function $\varphi$, this condition is exactly stating that the function $\varphi$ is an entire map from $\mathbb{R}^2$ to $\mathbb{R}$.

The situation for complete space-like surfaces is actually a lot nicer than what we presented up to now, as the representation as a graph is not only local but actually global as we prove in the proposition below.

**Proposition 2.1.** *Let $\Omega$ be a domain in $\mathbb{R}^2$ (that is, an open connected set), let $S \overset{\iota}{\hookrightarrow} \mathbb{M}^{2,1}$ be a space-like immersion and $\psi : \Omega \to S$ a parametrization of $S$. Suppose that $S$ is complete with respect to the induced Riemannian metric. Then $S$ is an entire graph and $\Omega$ is simply connected.*

Before laying out the proof let us recall a definition and a proposition from the theory of coverings. Let $M$ and $N$ two differentiable manifolds and $f : M \to N$ a smooth map.

**Definition 2.2.** We say that the map $f$ satisfies the path lifting property if for every $p \in N$ and for every path $\alpha : I \to M$ such that $\alpha(0) = f(p)$ there exists a path $\tilde{\alpha} : I \to N$ such that $\tilde{\alpha}(0) = p$ and $f \circ \tilde{\alpha} = \alpha$. We can assume without loss of generality that the paths involved are differentiable.

**Proposition 2.3.** *If $f$ is a local homeomorphism and satisfies the path lifting property, then it is a covering map.*

*Proof of Proposition 2.1.* The idea of the proof is to show that $\Pi \circ \iota$ is a global diffeomrphism. Clearly this implies that $\Pi$ restricted to $S$ is a diffeomeorphism on $\mathbb{R}^2$ and so that $S$ is an entire graph. From that we obtain also that $\Omega$ is simply connected as $\Pi \circ \iota \circ \psi$ is a diffeomorphism with $\mathbb{R}^2$.

Let us observe that the target of $\Pi \circ \iota$ is just $\mathbb{R}^2$, which is simply connected, and the domain is $\Omega$, which is connected since it is a domain. We already proved, by showing that $S$ is locally a graph, that the map $\Pi \circ \iota$ is a local homeomorphism.
Having met these conditions, if we show that $\Pi \circ \iota$ satisfies the path lifting property, we conclude that it is a smooth covering map whose domain is simply connected, hence a diffeomorphism.

Let us fix a point $z \in \Omega$ and a path $\gamma : [0,1] \to \mathbb{R}^2$ of bounded length, $\ell(\gamma) < \infty$, such that $\gamma(0) = (\Pi \circ \iota)(z)$. Our goal is to show that $\gamma$ admits a lift to a path $\gamma^*$ on $\Omega$ starting at $z$.
To this end, let us consider the set of times up to which such a lift exists,

$$\mathcal{J} := \{\tau \in [0,1] \mid \exists \gamma^* : [0,t] \to \Omega \text{ such that } (\Pi \circ \iota) \circ \gamma^* = \gamma \text{ and } \gamma^*(0) = z\},$$

and let us define $\bar{\tau} := \sup \mathcal{J}$.

Let us observe that $0 \in \mathcal{J}$ and that $\mathcal{J}$ is open, since $\Pi \circ \iota$ is a local homeomorphism.

If we prove that $\bar{\tau} = 1$, then $\gamma$ admits a lift on the whole interval $[0,1]$ and we have finished. Then let us suppose by contradiction that $\bar{\tau} < 1$ and define $\gamma^* : [0, \bar{\tau}) \to \Omega$ the lift of $\gamma$ up to $\bar{\tau}$. This curve exists by definition of $\bar{\tau}$.

Finally, we need to prove that the limit $\lim_{\tau \to \bar{\tau}} \gamma^*(t)$ exists. If this is true we can then extend $\gamma^*$ up to $\bar{\tau}$, which means that $\bar{\tau} \in \mathcal{J}$. Since $\mathcal{J}$ is open, there exists an element $\bar{\tau} + \varepsilon \in \mathcal{J}$, $\varepsilon > 0$. This contradicts the definition of $\bar{\tau}$ and

proves $\bar{\tau} = 1$.

To prove that $\lim_{\tau \to \bar{\tau}} \gamma^*(\tau)$ exists we just need a couple of observations: First we note that the map $\Pi$ dilates distances, as

$$\begin{aligned} \langle v, v \rangle_{\mathbb{M}^{2,1}} &= (v^1)^2 + (v^2)^2 - (v^3)^2 \\ &\leq (v^1)^2 + (v^2)^2 \\ &= \langle \Pi v, \Pi v \rangle_{\mathbb{R}^2} \, . \end{aligned}$$

From this it is clear that $\ell(\gamma^*) \leq \ell(\gamma)$.
Now, if we fix any sequence $\tau_k \to \bar{\tau}$ and observe that as $\gamma$ is differentiable, it is derivative is bounded, so $\ell(\gamma|_{[\tau_k, \tau_m]}) \leq C|\tau_k - \tau_m|$. By observing that

$$\begin{aligned} d(\gamma^*(\tau_k), \gamma^*(\tau_m)) &\leq \ell(\gamma^*|_{[\tau_k, \tau_m]}) \\ &\leq \ell(\gamma|_{[\tau_k, \tau_m]}) \\ &\leq C|\tau_k - \tau_m| \end{aligned}$$

we deduce that also $(\gamma^*(\tau_k))_k$ is a Cauchy sequence, hence converging since $S$ is complete and $\Omega$ is to be intended with the induced metric.
By the arbitrariness of the sequence $(\tau_k)_k$ we deduce the limit $\lim_{\tau \to \bar{\tau}} \gamma^*(\tau)$ exists. $\square$

In this section we have essentially proved that a complete space-like surface immersed in $\mathbb{M}^{2,1}$ is actually the graph of a smooth function $f$ satisfying $|\nabla f|^2 < 1$. A partial converse also holds: the graph of a smooth function whose gradient has norm strictly smaller than 1 is a space-like surface, although it may lack completeness.

## 2.3 Normal vectors to a space-like surface in $\mathbb{M}^{2,1}$

Since the ambient metric $g$ is nondegenerate and each tangent plane $T_pS$ is space-like, the restriction of $g$ to $T_pS$ is positive-definite, hence nondegenerate. It follows that, for every $p \in S$,

$$\mathbb{M}^{2,1} = T_p\mathbb{M}^{2,1} = (T_pS)^\perp \oplus T_pS \, .$$

Since $S$ has codimension 1 in $\mathbb{M}^{2,1}$, the orthogonal complement $(T_pS)^\perp$ is one-dimensional. Therefore it is generated by a vector $N = N_p$ normal to $S$.

We can easily show that $N$ must be a time-like vector, since $T_pS$ is space-like.
Let us fix $e_1, e_2 \in T_pM$ two independent vectors. Since $N$ belongs to $(T_pM)^\perp$ we have $\langle e_1, N \rangle = \langle e_2, N \rangle = 0$. This implies that when we look at the repre-

sentative matrix of $g_p$ written in the basis $(N, e_1, e_2)$, we have

$$[g_p] = \begin{pmatrix} \langle N, N\rangle & 0 & 0 \\ 0 & \langle e_1, e_1\rangle & \langle e_1, e_2\rangle \\ 0 & \langle e_2, e_1\rangle & \langle e_2, e_2\rangle. \end{pmatrix} =: \begin{pmatrix} \langle N, N\rangle & 0 \\ 0 & [(g|_S)_p] \end{pmatrix}.$$

Where the block $[(g|_S)_p]$ represents the form $g$ restricted to the subspace $T_pM$. We conclude that $\langle N, N\rangle$ is negative just by observing that $\det[g_p] < 0$ as $g$ is a Lorentzian metric, while $\det[(g|_S)_p] > 0$ since $g$ becomes a metric when restricted to $S$.

Since $(T_pS)^\perp$ is 1-dimensional, all of its vectors are proportional to each other. Moreover, as we have just seen, they are all timelike, so if $w$ is a nonzero vector in $(T_pS)^\perp$, then

$$\langle w, w\rangle < 0.$$

We can thus rescale $w$ by a factor $k \in \mathbb{R}$ such that $\langle kw, kw\rangle = -1$. This equation yields two possible choices of $k$, one for each connected component of the space $\hat{\mathbb{H}}^2 = \{v \in \mathbb{M}^{2,1} : \langle v, v\rangle = -1\}$. We can consistently choose the connected component with positive time

$$\mathbb{H} = \mathbb{H}^2 = \{v \in \mathbb{M}^{2,1} : \langle v, v\rangle = -1,\ v^3 > 0\} \quad ,$$

effectively selecting a unique normal vector for each $p \in S$. The choice of the component within $\{v^3 > 0\}$ or $\{v^3 < 0\}$ is sometimes referred to a "time-orientation" for the space $\mathbb{M}^{2,1}$, and can be performed independently from the surface $S$. This way we obtain a globally defined map

$$\tilde{u} : S \to \mathbb{H} \quad . \tag{2.1}$$

## 2.4 $\mathbb{H}$ is a model for the Hyperbolic plane

The space $\mathbb{H}$ that we just defined is clearly a surface immersed in $\mathbb{M}^{2,1}$, but there is more. First of all, we observe that, by manipulating the definition of $\mathbb{H}$, we can write it as the graph of the function $\varphi(x, y) = \sqrt{x^2 + y^2 + 1}$. The gradient of such function is $\nabla\varphi = \frac{1}{\sqrt{x^2+y^2+1}}(x, y)$, from which it is clear that $|\nabla\varphi| < 1$. By applying the result we proved before, we obtain that the bilinear form that $\mathbb{H}$ inherited from the immersion in $\mathbb{M}^{2,1}$ is a Riemannian metric. In fact this space is one of the many models of the hyperbolic plane.
Another model for the same Riemannian manifold is the Poincaré disk. As a differentiable manifold, the Poincaré disk is simply the unit ball in $\mathbb{R}^2$, $\mathbb{D} = \{\xi \in \mathbb{R}^2 \mid |\xi| \leq 1\}$, while we can construct the metric by exhibiting a diffeomorphism between $\mathbb{D}$ and $\mathbb{H}$ pulling back the metric.
A possible diffeomorphism is the map $p$, whose inverse is the map $q$, as can be

verified by direct computation:

$$\begin{aligned} p\colon \mathbb{H} &\longrightarrow \mathbb{D} & q\colon \mathbb{D} &\longrightarrow \mathbb{H} \\ (x,y,t) &\longmapsto \frac{1}{1+t}(x,y) & \xi &\longmapsto \frac{1}{1-|\xi|^2}(2\xi^1, 2\xi^2, 1+|\xi|^2). \end{aligned}$$

It is clear that both functions are smooth, since $|\xi| < 1$ and $t > 1$.

We would like to find the expression of $q^*(g_{\mathbb{H}})$.To simplify the calculations we might observe that the metric of $\mathbb{H}$ inherited by $\mathbb{M}^{2,1}$ is invariant under a rotation on the $t$ axis, and that such a rotation corresponds under $p$ to a rotation of the disk. This means that the metric $q^*(g_H)$ will be invariant under rotation.
It is then sufficient to fix a certain $\xi \in \mathbb{D}$ of the form $\xi = (a, 0)$ and calculate $q^*(g_{\mathbb{H}})_\xi$; this will specify the whole metric tensor. The matrix associated to $dq_\xi$ in the canonical bases $\mathscr{B}_1 = (e_1, e_2)$ of $\mathbb{R}^2$ and of $\mathscr{B}_2 = (E_1, E_2, E_3)$ of $\mathbb{M}^{2,1}$ is

$$[dq_\xi]_{\mathscr{B}_1\mathscr{B}_2} = \frac{2}{(1-|\xi|^2)^2} \begin{pmatrix} 1+(\xi^1)^2-(\xi^2)^2 & 2\xi^1\xi^2 \\ 2\xi^1\xi^2 & 1-(\xi^1)^2+(\xi^2)^2 \\ 2\xi^1 & 2\xi^2 \end{pmatrix}.$$

Which, evaluated at $\xi = (a, 0)$, yields

$$[dq_{(a,0)}]_{\mathscr{B}_1\mathscr{B}_2} = \frac{2}{1-a^2} \begin{pmatrix} \frac{1+a^2}{1-a^2} & 0 \\ 0 & 1 \\ \frac{2a}{1-a^2} & 0 \end{pmatrix} =: \frac{2}{1-a^2} \begin{pmatrix} v_1 | v_2 \end{pmatrix}.$$

We observe that, with respect to the metric of $\mathbb{M}^{2,1}$, it holds

$$\langle v_1, v_1 \rangle = 1, \quad \langle v_2, v_2 \rangle = 1, \quad \langle v_1, v_2 \rangle = 0.$$

Therefore $\mathscr{B}_3 = (v_1, v_2)$ is an orthonormal basis of $T_{q(0,a)}\mathbb{H}$ with respect to $g_{\mathbb{H}}$, hence $[g_{\mathbb{H}}]_{\mathscr{B}_3} = I$, while $dq_{(a,0)}$ represented by

$$[dq_{(a,0)}]_{\mathscr{B}_3\mathscr{B}_1} = \frac{2}{1-a^2} I.$$

So we have

$$\begin{aligned} [q^*(g_{\mathbb{H}})_{(0,a)}]_{\mathscr{B}_1} &= [dq_{(0,a)}]^T_{\mathscr{B}_3\mathscr{B}_1} [g_{\mathbb{H}}]_{\mathscr{B}_3} [dq_{(0,a)}]_{\mathscr{B}_3\mathscr{B}_1} \\ &= [dq_{(0,a)}]^T_{\mathscr{B}_3\mathscr{B}_1} I [dq_{(0,a)}]_{\mathscr{B}_3\mathscr{B}_1} \\ &= \left(\frac{2}{1-a^2}\right)^2 I \quad . \end{aligned}$$

In particular, $q^*(g_{\mathbb{H}})_{(0,a)}$ is a multiple of the Euclidean metric.

By the argument on rotational invariance, we conclude that for every $\xi \in \mathbb{D}$,

$$q^*(g_{\mathbb{H}})_\xi = \left(\frac{2}{1-|\xi|^2}\right)^2 g^E_\xi.$$

The disk $\mathbb{D}$ with this metric, which we denote also with $ds^2_p$, is called the Poincaré model of the Hyperbolic plane.

### 2.4.1 The Gauss Map in $\mathbb{M}^{2,1}$

In (2.1) we constructed a map $\tilde{u}$ between the surface $S$ and $\mathbb{H}$. Having now at our disposal the isometry $q$ between $\mathbb{H}$ and the Poincaré disk, $\mathbb{D}$, we can compose $\tilde{u}$ with $q$, obtaining what is called the Gauss map of the surface $S$.

$$u := q \circ \tilde{u} : S \to \mathbb{D}.$$

The construction we followed is analogous to the classical one, except that the target of the Gauss map is here $\mathbb{D}^2$ instead of $\mathbb{S}^2$. Moreover, unlike the Euclidean case, for a space-like surface we always have a globally defined unit normal field: once a time orientation of $\mathbb{M}^{2,1}$ is fixed, the timelike line $(T_pS)^\perp$ intersects the corresponding connected component of $\hat{\mathbb{H}}^2$ in exactly one point for every $p \in S$. Instead, for an immersed surface in Euclidean space, there is no canonical way to choose a normal vector.

# Chapter 3

# The Gauss-Codazzi equations and the curvatures of $S$

In this section we derive the Gauss-Codazzi equations for a space-like surface immersed in $\mathbb{M}^{2,1}$ and see that they assume a different form from the Euclidean case. These equations will play a central role in the discussion that follows, especially due to Bonnet's fundamental theorem on surfaces, stated at the end of the section.

## 3.1 Recall on the second fundamental form

We repeat now the construction of the second fundamental form of the immersion $\iota$, which brings a different sign convention that will occur also to the expression of the Gaussian curvature.

As we have already noted, given a point $p \in S$, since $S$ is a space-like surface, we can decompose the tangent space

$$T_p\mathbb{M}^{2,1} = T_pS \oplus (T_pS)^{\perp} \ .$$

This splitting is differentiable, in the sense that the mappings

$$(p, v) \mapsto (p, v^T) \qquad \text{and} \qquad (p, v) \mapsto (p, v^N)$$

are differentiable maps of $T\mathbb{M}^{2,1}|_S$ to $T\mathbb{M}^{2,1}|_S$.

Denote with $D$ the Levi-Civita connection on $\mathbb{M}^{2,1}$ induced by the pseudo-metric $g_{\mathbb{M}^{2,1}}$ and with $\nabla$ the Levi-Civita connection of $(S, \mathbf{I})$. As usual, if $X$ and $Y$ are local vector fields on $S$ with local extensions $\bar{X}$ and $\bar{Y}$ then the induced connection on $S$ is simply the projection of the connection of the ambient space onto the surface $S$

$$\nabla_X Y = \left(D_{\bar{X}}\bar{Y}\right)^T \ .$$

Which means that the difference between the two connections of $X$ and $Y$ will

be a local vector field on $\mathbb{M}^{2,1}$ normal to $S$,

$$B(X,Y)_p := (D_{\bar{X}}\bar{Y} - \nabla_X Y)_p \perp T_pS. \quad . \tag{3.1}$$

As in the Euclidean case, $B$ is a symmetric tensor and is independent of the chosen extensions $\overline{X}$ and $\overline{Y}$. Since the normal space $(T_pS)^\perp$ is one-dimensional, once a local unit normal vector field $N$ along $S$ is fixed, there exists a unique symmetric bilinear form $\mathbf{II} = \mathbf{II}(X,Y)$ such that

$$B(X,Y)_p = (\mathbf{II}(X,Y)N)_p,$$

From now on we forget to distinguish a vector field $X$ with its extension $\overline{X}$. Let $N$ be local normal vector field on $S$ and $V, W$ two local tangent vector fields to $S$. We observe first that

$$\langle N, W\rangle \equiv 0 \ ,$$

so that by taking the derivative along $V$,

$$\langle D_V N, W\rangle + \langle N, D_V W\rangle = 0 \quad .$$

Then

$$\begin{aligned}
\langle D_V N, W\rangle &= -\langle N, B(V,W) + \nabla_V W\rangle \\
&= -\langle N, B(V,W)\rangle \\
&= -\langle N, \mathbf{II}(V,W)N\rangle \\
&= -\langle N, N\rangle\, \mathbf{II}(V,W) \\
&= \mathbf{II}(V,W) \quad .
\end{aligned} \tag{3.2}$$

With this choice we have, from definition 3.1,

$$\boxed{D_V X = \nabla_V X + \mathbf{II}(V,X)N} \quad . \tag{3.3}$$

Now, having fixed $W_p \in T_pS$, and since the metric on $T_pS$ is a nondegenerate bilinear form, we can associate to the functional $\mathbf{II}(\cdot, W_p) : T_pS \to \mathbb{R}$ a unique vector $\mathbf{II}(\cdot, W_p)^\sharp \in T_pS$ such that for every $V_p \in T_pS$,

$$\left\langle \mathbf{II}(\cdot, W_p)^\sharp, V_p \right\rangle = \mathbf{II}(V_p, W_p) \ .$$

We then define the shape operator by

$$A_p W_p := \mathbf{II}(\cdot, W_p)^\sharp.$$

Thus

$$\begin{aligned}
\langle A_p W_p, V_p\rangle &= \mathbf{II}(V_p, W_p) \\
&= \mathbf{II}(W_p, V_p) \\
\text{[(3.2)]} \qquad &= \left\langle D_{W_p} N, V_p \right\rangle .
\end{aligned}$$

Since this holds for every $V_p \in T_pS$, we obtain

$$A_pW_p = D_{W_p}N.$$

So we have $A = DN$.

## 3.2 The Gauss-Codazzi equations and Gaussian curvature

It is easy to see that $\mathbb{M}^{2,1}$ is actually flat with its metric. This mean that its curvature tensor is zero $\overline{R}(X,Y)Z \equiv 0$. Let us rewrite $\overline{R}$ in relation with $R$ the curvature tensor of the surface $S$. By definition

$$\overline{R}(X,Y)Z = D_XD_YZ - D_YD_XZ - D_{[X,Y]}Z \quad .$$

Let us focus first on the term $D_XD_YZ$, applying the equation (3.3) we get

$$\begin{aligned} D_XD_YZ &= D_X(\nabla_YZ + \mathbf{II}(Z,Y)N) \\ &= \nabla_X\nabla_YZ + \mathbf{II}\left(\nabla_Y, X\right)N + d_X(\mathbf{II}(Z,Y))N + \mathbf{II}(Z,Y)D_XN \quad . \end{aligned}$$

And analogously

$$D_YD_XZ = \nabla_Y\nabla_XZ + \mathbf{II}\left(\nabla_X, Y\right)N + d_Y(\mathbf{II}(Z,X))N + \mathbf{II}(Z,X)D_YN \quad .$$

Finally we have, by noting that the connection $\nabla$ is symmetric,

$$\begin{aligned} D_{[X,Y]}Z &= \nabla_{[X,Y]}Z - \mathbf{II}([X,Y],Z)N \\ &= \nabla_{[X,Y]}Z + \mathbf{II}(Z,\nabla_XY)N - \mathbf{II}(Z,\nabla_YX)N \quad . \end{aligned}$$

So that, by putting everything together we obtain,

$$\begin{aligned} 0 &= \overline{R}(X,Y)Z \\ &= D_XD_YZ - D_YD_XZ - D_{[X,Y]}Z \\ &= \nabla_X\nabla_YZ - \nabla_Y\nabla_XZ - \nabla_{[X,Y]}Z \\ &\qquad + d_X(\mathbf{II}(Z,Y))N - \mathbf{II}(\nabla_XZ,Y)N - \mathbf{II}(Z,\nabla_XY)N \\ &\qquad - [d_Y(\mathbf{II}(Z,X))N - \mathbf{II}(\nabla_YZ,X)N - \mathbf{II}(Z,\nabla_YX)N] \\ &\qquad + \mathbf{II}(Z,Y)D_XN - \mathbf{II}(Z,X)D_YN \quad . \end{aligned} \tag{3.4}$$

Now, we recognize in the expression that $R(X,Y)Z = \nabla_X\nabla_YZ - \nabla_Y\nabla_XZ - \nabla_{[X,Y]}Z$, and by definition, we set the derivative of the tensor,

$$(\nabla_X(\mathbf{II}))(Y,Z) = d_X(\mathbf{II}(Z,Y)) - \mathbf{II}(\nabla_XZ,Y) - \mathbf{II}(Z,\nabla_XY) \ .$$

So, by continuing in (3.4),

$$0 = R(X,Y)Z + (\nabla_X \mathbf{II})(Y,Z)N - (\nabla_Y \mathbf{II})(Z,X)N \\ + \mathbf{II}(Z,Y)D_X N - \mathbf{II}(Z,X)D_Y N \quad . \tag{3.5}$$

Now we observe that from $\langle N, N\rangle \equiv -1$ we have, for $v$ tangent to $S$,

$$2\langle D_v N, N\rangle = 0$$

so that $D_v N$ is a tangent vector on $S$. Now, in (3.5) we can separate the terms that are tangent to $S$ and those that are normal. This way we obtain two distinct equations

$$\begin{cases} R(X,Y)Z = -\mathbf{II}(Z,Y)D_X N + \mathbf{II}(Z,X)D_Y N \\ (\nabla_X \mathbf{II})(Y,Z) - (\nabla_Y \mathbf{II})(Z,X) = 0 \end{cases} \quad . \tag{3.6}$$

These equations are (a form of) the Gauss-Codazzi equations.

Let us take the first equation of (3.5) and test it by a tangent local vector field $W$. We get

$$\begin{aligned} \langle R(X,Y)Z, W\rangle &= \langle -\mathbf{II}(Z,Y)D_X N + \mathbf{II}(Z,X)D_Y N, W\rangle \\ &= -\mathbf{II}(Z,Y)\langle D_X N, W\rangle + \mathbf{II}(Z,X)\langle D_Y N, W\rangle \\ &= -\mathbf{II}(Z,Y)\mathbf{II}(X,W) + \mathbf{II}(Z,X)\mathbf{II}(Y,W) \quad . \end{aligned}$$

Now, if we fix $v_1, v_2$ a basis of $T_pS$ and evaluate the last equation by setting $X = v_1, Y = v_2, Z = v_2, W = v_1$ and $h_{ij} = \mathbf{II}(v_i, v_j)$ we get

$$\begin{aligned} \langle R(v_1,v_2)v_2, v_1\rangle &= -\mathbf{II}(v_2,v_2)\mathbf{II}(v_1,v_1) + \mathbf{II}(v_2,v_1)\mathbf{II}(v_2,v_1) \\ &= -\left(\mathbf{II}(v_2,v_2)\mathbf{II}(v_1,v_1) - \mathbf{II}(v_1,v_2)^2\right) \\ &= -\det(h_{ij}) \quad . \end{aligned} \tag{3.7}$$

Now, by definition of sectional curvature and setting $g_{ij} = \mathbf{I}(v_i, v_j)$,

$$\langle R(v_1,v_2)v_2, v_1\rangle = K(v_1,v_2)\det(g_{ij}) \quad . \tag{3.8}$$

Here $K(v_1, v_2)$ is the sectional curvature of the plane spanned by $v_1, v_2$ which is simply the gaussian curvature $K$ since we're in dimension 2.
So by putting together (3.7) and (3.8),

$$K = -\det_{\mathbf{I}}(\mathbf{II}) = -\frac{\det(h_{ij})}{\det(g_{ij})} \quad .$$

Which is the same as the Gauss equation for surfaces in Euclidean space if not for a negative sign.
Let us also focus on the other equation. We can define the exterior derivative

of a tensor with respect to the connection,

$$d^{\nabla}T(X,Y) = \nabla_X(TY) - \nabla_Y(TX) - T[X,Y] = (\nabla_X T)(Y) - (\nabla_Y T)(X) \ .$$

This way, Codazzi's equation becomes

$$d^{\nabla}\mathbf{II} = 0.$$

So that the two equations get the following form:

$$\begin{cases} \det_{\mathbf{I}}(\mathbf{II}) = -K \\ d^{\nabla}\mathbf{II} = 0. \end{cases} \tag{3.9}$$

### 3.2.1 Bonnet's theorem

The Gauss-Codazzi equations play a fundamental role when the ambient manifold has constant sectional curvature. In this case, they are not only necessary compatibility conditions for the existence of an isometric immersion, but, on a simply connected surface, they are also sufficient.

**Theorem 3.1** (Bonnet)**.** *Let $(g_{ij})_{ij}$ and $(h_{ij})_{ij}$ be differentiable functions defined on an open set $V \subset \mathbb{R}^2$ to $\mathbb{R}$ such that $g = (g_{ij})_{ij}$ is is a symmetric positive definite matrix and $h = (h_{ij})_{ij}$ is symmetric. Suppose that they satisfy the equation* (3.9)*.*
*Then for every $q \in V$ there exists a neighborhood $U \subset V$ of $q$ and a diffeomorphism $\mathbf{x} : U \to \mathbf{x}(U) \subset \mathbb{M}^{2,1}$ such that the regular surface $\mathbf{x}(U)$ has $g$ and $h$ as first and second fundamental forms respectively.*
*Furthermore, if $U$ is connected and if*

$$\overline{\mathbf{x}} : U \to \overline{\mathbf{x}}(U) \subset \mathbb{M}^{2,1}$$

*is another diffeomorphism satisfying the same conditions, then there exist an isometry $T : \mathbb{M}^{2,1} \to \mathbb{M}^{2,1}$ such that $\overline{\mathbf{x}} = T \circ \mathbf{x}$.*

We redirect to [2] for a proof of the Euclidean case, whose only difference is the sign of Gauss' equation.

### 3.2.2 Mean curvature

We conclude this section by briefly recalling the definition of mean curvature, which does not differ from the usual Euclidean setting. As we have already seen, we can associate to the bilinear form $\mathbf{II}$ a linear operator $A_p : T_pS \to T_pS$, called shape operator, such that $\mathbf{II}(v,w) = \langle A_p(v), w\rangle$.
We define the mean curvature of $S$ at $p$ as

$$H(p) = \frac{1}{2}\operatorname{tr}(A_p) \ .$$

If we again fix a basis $v_1, v_2$, denote with $(g_{ij})_{ij}$ and $(h_{ij})_{ij}$ the coefficients of the first and second fundamental forms, and express $A_p$ as $A_p(v_i) = A_i{}^k v_k$, we get

$$h_{ij} = \mathbf{II}(v_i, v_j) = \langle A_p(v_i), v_j \rangle = A_i{}^k g_{kj} \quad .$$

If $(g^{ij})$ is the inverse matrix of $(g_{ij})$, it follows that

$$A_i{}^k = h_{ij} g^{jk},$$

and therefore

$$\mathrm{tr}(A_p) = A_i{}^i = h_{ij} g^{ji} = g^{ij} h_{ij} \quad .$$

So that we have

$$H = \frac{1}{2}\,\mathrm{tr}(A_p) = \frac{1}{2} A_i{}^i = \frac{1}{2} g^{ij} h_{ij} = \frac{1}{2}\,\mathrm{tr}_{\mathbf{I}}(\mathbf{II}) \quad .$$

Now, since

$$(g^{ij}) = \frac{1}{\det(g_{ij})} \begin{pmatrix} g_{22} & -g_{12} \\ -g_{12} & g_{11} \end{pmatrix} \quad ,$$

we obtain

$$2H = g^{ij} h_{ij} = \frac{h_{22} g_{11} - 2 h_{12} g_{12} + h_{11} g_{22}}{\det(g_{ij})} \quad .$$

As we already mentioned, there is no formal difference with surfaces immersed in $\mathbb{R}^3$.

# Chapter 4

# Riemann surfaces and harmonic maps

We here introduce some notation and some fundamental concepts of Riemann surfaces. The most crucial part will be the fact that each metric on a surface is locally conformal to the Euclidean, the relationship between a Riemannian metric and a complex structure, the Uniformization theorem and the definition of the Hopf differential.

## 4.1 Preliminaries on Riemann surfaces

Let us fix $S$ a surface (a topological manifold of dimension 2).

**Definition 4.1.** A complex atlas $\mathcal{C}$ is a collection of charts

$$\mathcal{C} = \{z_\alpha : U_\alpha \to \mathbb{C}\}$$

that covers the whole surface $S$ and whose transition maps

$$z_\beta \circ z_\alpha^{-1} : \ z_\alpha(U_\alpha \cap U_\beta) \to z_\beta(U_\alpha \cap U_\beta)$$

are holomorphic as maps between open subsets of $\mathbb{C}$. We call $\mathscr{C}$ a complex or conformal structure if it is an atlas maximal under the subset relation. Finally we call the pair $(S, \mathscr{C})$ (just $S$ for short) a Riemann surface.

**Definition 4.2.** A continuous map $h : S_1 \to S_2$ between Riemann surfaces is said to be holomorphic if, in local coordinates $\{U_\alpha, z_\alpha\}$ on $S_1$ and $\{U'_\beta, z'_\beta\}$ on $S_2$,all the maps $z'_\beta \circ h \circ z_\alpha^{-1}$ are holomorphic wherever they are defined. We call $f$ a biholomorphism if it is holomorphic, bijective and its inverse $f^{-1}$ is holomoprhic.
We say that the two Riemann surfaces $S_1$ and $S_2$ are isomorphic if there exists a biholomoprhism between them.

**Definition 4.3.** We say that two metrics $g$ and $g'$ on a surface $S$ are conformal

to each other if there exists a smooth function $\upsilon : S \to \mathbb{R}$ such that

$$g' = e^{2\upsilon} g \quad .$$

Equivalently, $g'$ is at each point a positive multiple of $g$ by a smooth function. The function $e^{2\upsilon}$ is called the conformal factor.

### 4.1.1 Conformal structures on 2-surfaces

Here we state the results that lead to the fact that every Riemannian surface admits a complex structure.

**Theorem 4.4.** *Given a Riemannian metric $g$ on a differentiable surface $S$, and a point $p \in S$, there is a chart near $p$, $\varphi : U \to \mathbb{R}^2$ such that $g$ takes the form and a smooth positive function $\lambda$ such that*

$$g = \lambda^2(dx^2 + dy^2) = e^{2\upsilon}(dx^2 + dy^2),$$

*where $(x, y)$ are the coordinates defined by $\varphi$.*

**Definition 4.5.** The chart $(\varphi, U)$ in the theorem is called conformal, while the coordinates $(x, y)$ on $U$ are called isothermal coordinates around $p$.

Given a conformal chart we may introduce the change of coordinates

$$\begin{cases} z = x + iy \\ \bar{z} = x - iy \end{cases} \quad \text{or equivalently} \quad \begin{cases} x = \frac{1}{2}(z + \bar{z}) \\ y = \frac{i}{2}(z - \bar{z}) \end{cases} \quad .$$

The associated coordinated fields and forms can be expressed as

$$\begin{cases} \dfrac{\partial}{\partial z} = \dfrac{1}{2}(\dfrac{\partial}{\partial x} - i\dfrac{\partial}{\partial y}) \\ \dfrac{\partial}{\partial \bar{z}} = \dfrac{1}{2}(\dfrac{\partial}{\partial x} + i\dfrac{\partial}{\partial y}) \end{cases} \quad \text{and} \quad \begin{cases} dz = dx + idy \\ d\bar{z} = dx - idy \end{cases} \quad .$$

This lets us write the Euclidean metric as

$$\begin{aligned} dx^2 + dy^2 &= dx \otimes dx + dy \otimes dy \\ &= \frac{1}{4} d(z + \bar{z}) \otimes d(z + \bar{z}) - \frac{1}{4} d(z - \bar{z}) \otimes d(z - \bar{z}) \\ &= \frac{1}{2}(dz \otimes d\bar{z} + d\bar{z} \otimes dz) \\ &= \Re(dz \otimes d\bar{z}) \\ &= dz d\bar{z} \\ &= |dz|^2 \quad , \end{aligned}$$

where the last two equalities are definitions.

So we can express the metric $g$ as

$$g = e^{2v}|dz|^2.$$

The change of coordinates $(x, y) \mapsto z = x + iy$ yields charts with values in $\mathbb{C}$. Using these charts, we can endow $S$ with a complex structure.

**Proposition 4.6.** *Fix an orientation on $S$. Given a conformal chart $(\varphi, U)$ preserving the orientation, denote by $(\tilde{\varphi}, U)$ the associated complex chart. Then*

$$\{(\tilde{\varphi}, U) \mid (\varphi, U) \text{ is an orientation-preserving conformal chart on } (S, g)\}$$

*constitutes a complex structure on $S$.*

Now, let us recall some operators that take a simple form when read through a conformal chart, for which the metric has the form $e^{2u}|dz|^2$. The proof of these formulas can be found in [7].
First, the Laplacian with respect to the metric $g$ is

$$\Delta_g = \frac{1}{\lambda^2}(\frac{\partial}{\partial x^2} + \frac{\partial}{\partial y^2}) = e^{-2u}(\frac{\partial}{\partial x^2} + \frac{\partial}{\partial y^2}) = e^{-2u}\frac{\partial}{\partial z}\frac{\partial}{\partial \bar{z}}.$$

While the Gauss curvature of the surface takes the form

$$K_g = -\Delta_g \log(\lambda) = -2\Delta u.$$

Let us finally observe that,

$$\begin{aligned} dz^2 &= dz \otimes dz \\ &= d(x + iy) \otimes d(x + iy) \\ &= dx \otimes dx + i\,(dx \otimes dy + dy \otimes dx) - dy \otimes dy \\ &=: dx^2 - dy^2 + 2idxdy\ , \end{aligned}$$

where we define $dxdy = \frac{1}{2}(dx \otimes dy + dy \otimes dx)$.

### 4.1.2 Integration through a conformal chart

Given a curve $\gamma$ in S, its length is defined by

$$\ell(\gamma) = \int_\gamma \lambda(z)|dz|\ .$$

On the other hand the area form can be expressed as follows

$$\lambda^2 dx \wedge dy = -\frac{i\lambda^2}{2} dz \wedge d\bar{z}\ ,$$

so that the area of a measurable set becomes

$$\begin{aligned}\mathrm{Area}(B) &:= \int_B \lambda^2(z)\frac{i}{2}dz \wedge d\bar{z} \\ &= \int_B \lambda^2(z)\frac{i}{2}dzd\bar{z} \quad .\end{aligned}$$

## 4.2 Uniformization

An important property of Riemann surfaces is that they can be classified in just three categories, depending on their universal covering. This result is (a direct consequence of) the uniformization theorem. In what follows we denote by $\widehat{\mathbb{C}}$ the 2-sphere, with the metric and conformal structure induced by the Euclidean metric of $\mathbb{R}^3$.

**Theorem 4.7** (Uniformization Theorem)**.** *Let $\Sigma$ be a simply connected Riemann surface. Then $\Sigma$ is isomorphic to exactly one of the following three surfaces:*

1. *The Riemann sphere $\widehat{\mathbb{C}}$.*
2. *The complex plane $\mathbb{C}$.*
3. *The Poincaré disk $\mathbb{D}$.*

Every Riemann surface $\Sigma$ admits a universal covering $\tilde{\Sigma}$, which inherits the complex structure of its base space. Since the universal covering $\tilde{\Sigma}$ of a surface $\Sigma$ is simply connected, we can apply the uniformization theorem to it, making $\Sigma$ fall in one of the following categories. We say that $\Sigma$ is

- elliptic if $\widetilde{\Sigma} \cong \widehat{\mathbb{C}}$,
- parabolic if $\widetilde{\Sigma} \cong \mathbb{C}$,
- hyperbolic if $\widetilde{\Sigma} \cong \mathbb{D}$.

In our case the surface we consider is simply connected and noncompact (for instance because it is the graph of an entire function on $\mathbb{R}^2$) so we can exclude the elliptic case.

**Corollary 4.8.** *If $\Sigma$ is a simply connected noncompact Riemann surface, then $\Sigma$ is conformally equivalent either to $\mathbb{C}$ or to $\mathbb{D}$.*

## 4.3 Harmonic maps between Riemann surfaces

Let $(\Sigma_1, g)$ and $(\Sigma_2, h)$ be two Riemann surfaces, and let $z$ and $w$ be local conformal coordinates on $\Sigma_1$ and $\Sigma_2$, respectively, so that the metrics are written as

$$g = e^{2\eta}|dz|^2 = \lambda^2|dz|^2, \qquad h = \varrho^2(w)|dw|^2 \quad .$$

Let $f : \Sigma_1 \to \Sigma_2$ be a smooth map, and denote for simplicity by $w$ its local expression in the chosen coordinates. In these coordinates the energy density of $f$ is given by

$$\begin{aligned} e(f) &= \frac{\varrho^2(w)}{\lambda^2}(|w_x|^2 + |w_y|^2) \\ &= 2\frac{\varrho^2(w)}{\lambda^2}(|w_z|^2 + |w_{\bar{z}}|^2) \quad . \end{aligned}$$

Then, if $\Omega \subset \Sigma_1$ is a compact domain, the energy of $f$ on $\Omega$ is

$$\begin{aligned} E_\Omega(f) &:= \int_\Omega e(f)\, dA_g \\ &= \int_\Omega \lambda^2(z) e(f(z))\, \frac{i}{2}\, dz d\bar{z} \\ &= \int_\Omega \varrho^2(w)(|w_z|^2 + |w_{\bar{z}}|^2)\, i\, dz d\bar{z} \quad . \end{aligned}$$

We say that $f$ is harmonic if it is a critical point of this functional with respect to compactly supported perturbations. The corresponding Euler-Lagrange equation is

$$w_{z\bar{z}} + 2\frac{\varrho_w}{\varrho} w_z w_{\bar{z}} = 0 \quad .$$

For a complete derivation of the compact case we redirect the reader to [7].

An important observation is that the local harmonic map equation does not involve the conformal factor $\lambda$ of the metric on the domain. This reflects the fact that, in dimension two, harmonicity depends only on the conformal class of the metric on the source surface.

### 4.3.1 Energy densities and Jacobian

In addition to the energy density of the map $f$, it is convenient to introduce two more refined quantities, the holomorphic and anti-holomorphic energy densities. These objects will let us also express the Jacobian of $f$ in a convenient way.

**Definition 4.9.** We define the holomorphic and antiholomorphic energy density of a smooth map $f : \Sigma_1 \to \Sigma_2$ respectively as

$$\mathcal{H}(f) := \frac{\varrho^2(f)}{\lambda^2}|f_z|^2, \qquad \text{and} \qquad \mathcal{L}(f) := \frac{\varrho^2(f)}{\lambda^2}|f_{\bar{z}}|^2 \quad .$$

It is immediate to note that

$$e(f) = \mathcal{H}(f) + \mathcal{L}(f) \ .$$

The Jacobian of the map is by definition

$$J(f) = \frac{\varrho^2(f)}{\lambda^2}(|f_z|^2 - |f_{\bar{z}}|^2) \quad .$$

So that we also have

$$J(f) = \mathcal{H}(f) - \mathcal{L}(f) \ . \ . \tag{4.1}$$

## 4.4 Harmonic maps into $\mathbb{H}$

We now specialize the previous discussion to maps with values in the hyperbolic plane. Since $\mathbb{H}$ is itself a Riemann surface, the notion of harmonic map introduced above already applies to maps with target $\mathbb{H}$. Nevertheless, in order to relate harmonicity to the geometry of a space-like surface in $\mathbb{M}^{2,1}$, it is convenient to rewrite the first variation in an extrinsic way, regarding $\mathbb{H}$ as the hyperboloid in Minkowski space.

Let us consider

$$f : (\mathbb{D}, e^{2\eta}(dx^2 + dy^2)) \to \mathbb{H} \subset \mathbb{M}^{2,1} \quad .$$

Since the metric on $\mathbb{H}$ is the one induced by the Minkowski product of $\mathbb{M}^{2,1}$, the energy density can be written as

$$e(f) = e^{-2\eta}(f_x \cdot f_x + f_y \cdot f_y) \quad .$$

When the energy density is integrable, we define the energy of the map by

$$\begin{aligned} E(f) &= \int_{\mathbb{D}} e^{2\eta} e(f) dx \wedge dy \\ &= \int_{\mathbb{D}} f_x \cdot f_x + f_y \cdot f_y dx \wedge dy \quad . \end{aligned}$$

As before, this quantity may well be infinite for a general map. However, if we consider compactly supported deformations, the corresponding first variation is still well defined, and this is enough for our purposes.

Let us fix a compactly supported deformation of $f$, namely a smooth map

$$\xi : \mathbb{D} \times (-\varepsilon, \varepsilon) \to \mathbb{H}$$

such that

$$\xi(x, y, 0) = f(x, y)$$

for every $(x, y) \in \mathbb{D}$, and there exists a compact set $K \subset \mathbb{D}$ such that

$$\xi(x, y, t) = f(x, y)$$

for every $(x, y) \in \mathbb{D} \setminus K$ and every $t \in (-\varepsilon, \varepsilon)$. We call $\xi$ of this form an admissible deformation of $f$.
Then, we define

$$f_t := \xi(\cdot, \cdot, t) \quad .$$

If $\Omega \subset \mathbb{D}$ is a compact domain containing $K$ in its interior, we define

$$\Delta E(t) := \int_\Omega e^{2\eta}(e(f_t) - e(f))\, dx \wedge dy \quad .$$

Since $f_t = f$ on $\mathbb{D} \setminus K$, the quantity $\Delta E(t)$ is independent of the choice of $\Omega$, provided $\Omega$ is sufficiently large.

**Definition 4.10.** We say that $f$ is a harmonic map if for every admissible deformation $\xi$ one has

$$\left.\frac{dE(f_t)}{dt}\right|_{t=0} := \lim_{t\to 0} \frac{\Delta E(t)}{t} = 0 \quad .$$

(We use the notation $\left.\frac{dE(f_t)}{dt}\right|_{t=0}$ for simplicity).

The following lemma is needed to give later a characterizaion of harmonic maps.

**Lemma 4.11.**

$$\left.\frac{dE(f_t)}{dt}\right|_{t=0} = -2\int_{\mathbb{D}} \xi_t \cdot (f_{xx} + f_{yy}) dx \wedge dy \ .$$

*Proof.* First we observe that

$$\begin{aligned}\left.\frac{dE(f_t)}{dt}\right|_{t=0} &= \int_{\mathbb{D}} \left.\frac{\partial}{\partial t}(\xi_x \cdot \xi_x + \xi_y \cdot \xi_y)\right|_{t=0} dx \wedge dy \\ &= 2\int_{\mathbb{D}} (\xi_x \cdot \xi_{xt} + \xi_y \cdot \xi_{yt})\big|_{t=0}\, dx \wedge dy \quad . \end{aligned} \tag{4.2}$$

Now,

$$\begin{aligned} d(\xi_t \cdot \xi_x\, dy) &= (\xi_t \cdot \xi_x)_x\, dx \wedge dy \\ &= (\xi_{xt} \cdot \xi_x + \xi_t \cdot \xi_{xx})\, dx \wedge dy, \end{aligned}$$

so that

$$\xi_x \cdot \xi_{xt}\, dx \wedge dy = d(\xi_t \cdot \xi_x\, dy) - \xi_t \cdot \xi_{xx}\, dx \wedge dy$$

and similarly

$$\xi_y \cdot \xi_{yt}\, dx \wedge dy = -d(\xi_t \cdot \xi_y\, dx) - \xi_t \cdot \xi_{yy}\, dx \wedge dy \ .$$

Now, by substituting into (4.2), we obtain

$$\begin{aligned}\left.\frac{dE(f_t)}{dt}\right|_{t=0} &= -2\int_{\mathbb{D}} \xi_t \cdot (\xi_{xx} + \xi_{yy})\big|_{t=0}\, dx \wedge dy \\ &\quad + 2\int_{\mathbb{D}} d(\xi_t \cdot \xi_x\, dy - \xi_t \cdot \xi_y\, dx)\big|_{t=0} \quad . \end{aligned}$$

Then, since $\xi_t$ has compact support $K$, by applying Stokes' theorem we obtain that the second term of the sum in the last equation is zero. Once recognized that $\xi_{xx}|_{t=0} = f_{xx}$ and $\xi_{yy}|_{t=0} = f_{yy}$ we have proven the lemma. $\square$

**Corollary 4.12.** *Let $f : \mathbb{D} \to \mathbb{H} \subset \mathbb{M}^{2,1}$ a smooth map. Then $f$ is harmonic if and only if the tangent component of its Laplacian is null:*

$$(\Delta f)^T = 0 \ .$$

*That is,*

$$\Delta f + \langle \Delta f, f \rangle \, f = 0 \ .$$

*Proof.* Let us fix $\xi : \mathbb{D} \times (-\varepsilon, \varepsilon) \to \mathbb{H}$ a compactly supported deformation of $f$ (that is, a deformation of $f$ as above). By lemma 4.11,

$$\left.\frac{dE(f_t)}{dt}\right|_{t=0} = -2 \int_{\mathbb{D}} \xi_t|_{t=0} \cdot \Delta f \ .$$

Being $\xi$ a deformation of $f$, which is a smooth map with values in $\mathbb{H}$, we have constantly $\xi(x, y, t) \in \mathbb{H}$, hence $\xi_t(x, y)|_{t=0} \in T_{\xi(x,y,0)}\mathbb{H}^2$. So we obtain

$$\left.\frac{dE(f_t)}{dt}\right|_{t=0} = -2 \int_{\mathbb{D}} \xi_t \cdot (\Delta f)^T \ . \tag{4.3}$$

Now, one implication is trivial, if $(\Delta f)^T = 0$, then also the LHS is zero for every choice of $\xi$.

On the other hand, let us assume that

$$\left.\frac{dE(f_t)}{dt}\right|_{t=0} = 0$$

for every admissible deformation and prove that $(\Delta f)^T$ is zero. We may suppose by contradiction $(\Delta f)^T \neq 0$ at some point $p$, this implies it is nonzero in a small neighborhood $U$ of $p$.
Then we choose a smooth function $\psi \geq 0$ of compact support contained in $U$, such that $\psi(p) > 0$, and define the deformation

$$\xi(x, y, t) = \exp_{f(x,y)}\big(t\psi(x, y)(\Delta f)^T(x, y)\big)$$

By substituting $\xi$ in the expression (4.3) we obtain

$$\left.\frac{dE(f_t)}{dt}\right|_{t=0} = -2 \int_{\operatorname{supp}\psi} \psi \, |(\Delta f)^T|^2 dx \wedge dy < 0 \ ,$$

where we have the strict inequality since we have both $\psi > 0$ and $(\Delta f)^T \neq 0$ on $\operatorname{supp}\psi \subset U$. This is absurd because we showed that the leftmost side of the last expression is exactly zero.

□

Let us fix now a conformal parametrization of our surface,

$$\sigma : \mathbb{D} \to \mathbb{M}^{2,1},$$

and let

$$N : \mathbb{D} \to \mathbb{M}^{2,1}$$

be the gauss map of the surface. We want to prove the following theorem:

**Theorem 4.13.**

$$(\Delta N)^T = 2(\nabla H) \ ,$$

*where $H$ is the mean curvature of the surface.*

In order to prove it, we need a lemma.

**Lemma 4.14.** *Let $B$ a 1-1 tensor on a Riemannian surface $M$, $v \in T_pM$, and denote with $\nabla$ the Levi-Civita connection of $M$. Then*

$$\operatorname{tr}(\nabla_v B) = d(\operatorname{tr} B)(v) \ .$$

*Proof.* Let us suppose the dimension of the manifold to be 2 for simplicity. Let $(e_i)_i$ be an orthonormal frame around $p$.
Let us observe

$$\begin{aligned} d\operatorname{tr}(B)(v) &= v\left(\sum_i \langle B(e_i), e_i\rangle\right) \\ &= \sum_i \langle \nabla_v(B(e_i)), e_i\rangle + \sum_i \langle B(e_i), \nabla_v e_i\rangle \\ &= \sum_i \langle (\nabla_v B)(e_i), e_i\rangle + \sum_i [\langle B(\nabla_v e_i), e_i\rangle + \langle B(e_i), \nabla_v e_i\rangle] \\ &= \operatorname{tr}(\nabla_v B) + \sum_i [\langle B(\nabla_v e_i), e_i\rangle + \langle B(e_i), \nabla_v e_i\rangle] \quad . \end{aligned} \tag{4.4}$$

So that we have our thesis if we show that the second term of the expression in (4.4) is zero. Suppose

$$\nabla_v e_1 = a e_1 + b e_2, \qquad \nabla_v e_2 = c e_1 + d e_2 \ . \ ,$$

for certain coefficients $a, b, c, d \in \mathbb{R}$.
Observe then that

$$\begin{aligned} \langle e_i, e_j\rangle &= \delta_{ij} \\ \Rightarrow \quad v\left(\langle e_i, e_j\rangle\right) &= v(\delta_{ij}) = 0 \\ \Rightarrow \quad \langle \nabla_v e_i, e_j\rangle + \langle e_i, \nabla_v e_j\rangle &= 0 \\ \Rightarrow \quad \langle \nabla_v e_i, e_j\rangle &= -\langle e_i, \nabla_v e_j\rangle \, . \end{aligned}$$

By setting $i = j$ we obtain $\langle \nabla_v e_1, e_1\rangle = a = \langle \nabla_v e_2, e_2\rangle = d = 0$ and if $i \neq j$ we get $\langle \nabla_v e_1, e_2\rangle = b = -\langle \nabla_v e_2, e_1\rangle = -c$ . Hence

$$\nabla_v e_1 = b e_2, \qquad \nabla_v e_2 = -b e_1.$$

Now we can rewrite the second term of (4.4),

$$\begin{aligned}\langle B(\nabla_v e_1), e_1\rangle + \langle B(e_1), \nabla_v e_1\rangle + \langle B(\nabla_v e_2), e_2\rangle + \langle B(e_2), \nabla_v e_2\rangle = \\ b\,\langle B(e_2), e_1\rangle + b\,\langle B(e_1), e_2\rangle - b\,\langle B(e_1), e_2\rangle - b\,\langle B(e_2), e_1\rangle = 0 \quad .\end{aligned}$$

Which proves the lemma. □

We are ready to prove theorem 4.13.

*Proof of theorem 4.13.* First we recall that if $A$ is the shape operator, then $N_x = A\partial_x$ and $N_y = A\partial_y$. If we assume that $\mathbf{I} = e^{2\eta}(dx^2 + dy^2)$ then $e^{-\eta}\partial_x$, $e^{-\eta}\partial_y$ is an orthonormal frame.
By applying the covariant derivatives along $\partial_x$ and $\partial_y$ to $N_x$ and $N_y$, respectively, we obtain

$$\begin{aligned}N^T_{xx} &= (\nabla_{\partial_x} A)(\partial_x) + A(\nabla_{\partial_x}\partial_x) \\ N^T_{yy} &= (\nabla_{\partial_y} A)(\partial_y) + A(\nabla_{\partial_y}\partial_y) \quad .\end{aligned}$$

So we obtain

$$\begin{aligned}(\Delta_{\mathbf{I}} N)^T = e^{-2\eta}\big((\nabla_{\partial_x} A)(\partial_x) + (\nabla_{\partial_y} A)(\partial_y)\big) \\ + e^{-2\eta} A(\nabla_{\partial_x}\partial_x + \nabla_{\partial_y}\partial_y) \quad .\end{aligned} \tag{4.5}$$

Now let us show that $\nabla_{\partial_x}\partial_x + \nabla_{\partial_y}\partial_y = 0$.
First, observe that since $\partial_x, \partial_y$ are orthogonal, $\langle \partial_x, \partial_y\rangle = 0$ we have $\left\langle \nabla_{\partial_y}\partial_x, \partial_y\right\rangle = -\left\langle \partial_x, \nabla_{\partial_y}\partial_y\right\rangle$. Furthermore, from the expression of the metric we have $\langle \partial_x, \partial_x\rangle = e^{2\eta} = \langle \partial_y, \partial_y\rangle$ so that $\partial_y(\langle \partial_x, \partial_x\rangle) = \partial_x(\langle \partial_y, \partial_y\rangle)$ and also, the connection is symmetric so that $\nabla_{\partial_x}\partial_y = \nabla_{\partial_y}\partial_x$. Then,

$$\begin{aligned}\langle \nabla_{\partial_x}\partial_x, \partial_y\rangle &= -\langle \partial_x, \nabla_{\partial_x}\partial_y\rangle \\ &= -\frac{1}{2}\partial_y\,\langle \partial_x, \partial_x\rangle \\ &= -\frac{1}{2}\partial_y\,\langle \partial_y, \partial_y\rangle \\ &= -\left\langle \nabla_{\partial_y}\partial_y, \partial_y\right\rangle \quad ,\end{aligned} \qquad \begin{aligned}\langle \nabla_{\partial_x}\partial_x, \partial_x\rangle &= \frac{1}{2}\partial_x\,\langle \partial_x, \partial_x\rangle \\ &= \frac{1}{2}\partial_x\,\langle \partial_y, \partial_y\rangle \\ &= \langle \nabla_{\partial_x}\partial_y, \partial_y\rangle \\ &= \left\langle \nabla_{\partial_y}\partial_x, \partial_y\right\rangle \\ &= -\left\langle \partial_x, \nabla_{\partial_y}\partial_y\right\rangle \quad .\end{aligned}$$

At this point, by developing on the base $\partial_x, \partial_y$ it is clear that

$$\nabla_{\partial_x}\partial_x + \nabla_{\partial_y}\partial_y = 0 \ .$$

So in (4.5) we obtain

$$(\Delta N)^T = e^{-2\eta}\big((\nabla_{\partial_x} A)(\partial_x) + (\nabla_{\partial_y} A)(\partial_y)\big) \ . \tag{4.6}$$

Since the shape operator $A$ is defined by $\mathbf{II}(Y, Z) = \langle AY, Z\rangle$, we can relate the

covariant derivatives of $A$ and $\mathbf{II}$ as follows:

$$\begin{aligned}(\nabla_X \mathbf{II})(Y,Z) &= X(\mathbf{II}(Y,Z)) - \mathbf{II}(\nabla_X Y, Z) - \mathbf{II}(Y, \nabla_X Z)\\ &= X(\langle AY, Z\rangle) - \langle A(\nabla_X Y), Z\rangle - \langle AY, \nabla_X Z\rangle\\ &= \langle \nabla_X(AY), Z\rangle + \cancel{\langle AY, \nabla_X Z\rangle} - \langle A(\nabla_X Y), Z\rangle - \cancel{\langle AY, \nabla_X Z\rangle}\\ &= \langle (\nabla_X A)(Y), Z\rangle \quad .\end{aligned}$$

Then we can apply the Codazzi equation in the following form:

$$\begin{aligned}\Big\langle (\nabla_{\partial_y} A)(\partial_y), \partial_x \Big\rangle &= (\nabla_{\partial_y} \mathbf{II})(\partial_y, \partial_x)\\ &= (\nabla_{\partial_y} \mathbf{II})(\partial_x, \partial_y)\\ &= \langle (\nabla_{\partial_x} A)(\partial_y), \partial_y\rangle \quad . \end{aligned} \tag{4.7}$$

So, finally, by taking the scalar product of (4.6) with $\partial_x$, we obtain

$$\begin{aligned}\Big\langle (\Delta N)^T, \partial_x \Big\rangle &= e^{-2\eta}\left[\langle (\nabla_{\partial_x} A)(\partial_x), \partial_x\rangle + \Big\langle (\nabla_{\partial_y} A)(\partial_y), \partial_x \Big\rangle\right]\\ \text{[(4.7)]} \qquad &= e^{-2\eta}\left[\langle (\nabla_{\partial_x} A)(\partial_x), \partial_x\rangle + \langle (\nabla_{\partial_x} A)(\partial_y), \partial_y\rangle\right]\\ &= \big\langle (\nabla_{\partial_x} A)(e^{-\eta}\partial_x), e^{-\eta}\partial_x\big\rangle + \big\langle (\nabla_{\partial_x} A)(e^{-\eta}\partial_y), e^{-\eta}\partial_y\big\rangle\\ &= \operatorname{tr}(\nabla_{\partial_x} A)\\ \text{[Lemma 4.14]} \qquad &= d\operatorname{tr}(A)(\partial_x)\\ &= 2\partial_x(H) = 2H_x \ .\end{aligned}$$

Analogously we can show that

$$\Big\langle (\Delta N)^T, \partial_y \Big\rangle = 2H_y \quad .$$

Therefore we have the thesis. □

We have now all the pieces needed to prove the central theorem of this subsection

**Theorem 4.15.** *Let $\sigma : (\mathbb{D}, e^{2\eta}(dx^2 + dy^2)) \to S \subset M$ a parametrization of a space-like surface and denote with $N : \mathbb{D} \to \mathbb{M}^{2,1}$ the corresponding Gauss map. Then $S$ has constant mean curvature if and only if $N$ is harmonic.*

*Proof.* Just observe that for theorem 4.13 we have

$$(\Delta N)^T = 2\nabla H.$$

Then recall from corollary 4.12 that $N$ if and only if the LHS in the last equation is zero, which in turn is equivalent to $H$ being constant. □

## 4.5 Energy densities and the Hopf differential

### 4.5.1 Quadratic differentials

To each map between Riemann surfaces we can associate a quantity that carries information about how far the map is from being conformal.

**Definition 4.16.** Given $\Sigma$ a Riemann surface we say that a symmetric 0-2 tensor field on $\Sigma$ with values in $\mathbb{C}$ is compatible with the complex structure if it can be expressed locally in isothermal coordinates as $\Phi = \varphi(z)dz^2$. We say that $\Phi$ is holomorphic if $\varphi$ is a holomorphic function.

It is clear that if we ha a change of coordiante $z \mapsto w = \gamma(z)$, then we can express the differentials as $dw = \gamma'(z)dz$. This means that the tensor product $dz^2$ will transform as

$$dw^2 = \gamma'(z)^2 dz^2.$$

As it is usual the case, a set of local representations of holomoprhic quadratic differentials $(dw_\alpha^2)_\alpha$ determines a holomorphic quadratic differential on the whole surface if they satisfy the transformation rule above.

To each map between Riemann surfaces, we can associate a quadratic differential, which gives us in some sense a characterization of harmonic local diffeomorphisms

**Lemma 4.17.** *Let $f : \Sigma_1 \to \Sigma_2$ a harmonic map between Riemann surfaces, with $\Sigma_1$ being endowed with a metric of nonpositive curvature. Let $z$ be a conformal parameter on $\Sigma_1$. Then if $w$ is a local representation of $f$,*

$$q = \varrho^2(w) w_z w_{\overline{z}} dz^2$$

*is a holomorphic quadratic differential on $\Sigma_1$.*

*Proof.* Let us show first that the expression of $q$ doesn't change under conformal transformations of the domain and the target.

First let us consider a conformal transformation on the domain, $z = \psi(z')$. Then $dz = \psi'(z')dz'$, $dz^2 = (\psi(z'))^2(dz')^2$,

$$w_{z'} = w_z \psi'(z'), \qquad \text{and} \qquad \overline{w}_{z'} = \overline{w}_z \psi'(z') \ .$$

So the metric transforms as

$$\begin{aligned} \varrho^2(w) w_{z'} \overline{w}_{z'} &= \varrho^2(w) w_z \overline{w}_z (\psi'(z'))^2 (dz')^2 \\ &= \varrho^2(w) w_z \overline{w}_z dz^2 \ . \end{aligned}$$

Hence $q$ changes as

$$\begin{aligned} q = \varrho^2(w) w_z \overline{w}_z dz^2 &= \varrho^2(w) w_z \overline{w}_z (\psi'(z'))^2 (dz')^2 \\ &= \varrho^2(w) w_z \overline{w}_z dz^2 \ . \end{aligned}$$

Then we consider a conformal transformation of the target, $w = \phi(w')$. As before we have $dw = \phi'(w')dw'$, and

$$\begin{aligned}\varrho^2(w)|dw|^2 &= \varrho^2(\phi(w'))|\phi'(w')|^2|dw'|^2 \\ &= \varrho^2(w')|dw'|^2 \ .\end{aligned}$$

Then

$$\begin{aligned}q &= \varrho^2(w)w_z\overline{w}_z dz^2 \\ &= \varrho^2(\phi(w'))(\phi'(w')w'_z\overline{\phi'(w')}\overline{w}_z)dz^2 \\ &= \varrho^2(\phi(w'))(|\phi'(w)|^2 w_z\overline{w}_z dz^2 \\ &= \varrho^2(w')w_z\overline{w}_z dz^2 \ .\end{aligned}$$

Finally, if $f$ is harmonic, we may show that $\varrho^2(w)w_z\overline{w}_z$ is holomorphic

$$\begin{aligned}\frac{\partial}{\partial\overline{z}}(\varrho^2(w)w_z\overline{w}_z) &= \frac{\partial}{\partial\overline{z}}(\varrho^2(w))w_z w_{\overline{z}} + \varrho^2(w)w_{z\overline{z}}\overline{w}_z + \varrho^2(w)w_z\overline{w}_{z\overline{z}} \\ &= 2\varrho(w)(\varrho_w(w)w_{\overline{z}} + \varrho_{\overline{w}}(w)\overline{w}_{\overline{z}})w_z\overline{w}_z + \varrho^2(w)w_{z\overline{z}}\overline{w}_z + \varrho^2(w)w_z\overline{w}_{z\overline{z}} \\ &= \overline{w}_z(\varrho^2(w)w_{z\overline{z}} + 2\varrho(w)\varrho_w(w)w_{\overline{z}}w_z)+ \\ &\quad + \overline{w}_z(\varrho^2(\overline{w})\overline{w}_{z\overline{z}} + 2\varrho(\overline{w})\varrho_{\overline{w}}(\overline{w})\overline{w}_{\overline{z}}\overline{w}_z) \\ &= \overline{w}_z(\varrho^2(w)w_{z\overline{z}} + 2\varrho(w)\varrho_w(w)w_{\overline{z}}w_z)+ \\ &\quad + \overline{w}_z\overline{(\varrho^2(w)w_{z\overline{z}} + 2\varrho(w)\varrho_w(w)w_{\overline{z}}w_z)} = 0\end{aligned}$$

,

where the last equality holds by applying the local equation for harmonicity. □

We can actually give a decomposition of any symmetric 0-2 tensor field

**Lemma 4.18.** *Let $\omega$ a symmetric 0-2 tensor field on $(\Sigma, g)$ with values in $\mathbb{C}$ . Then $\omega$ can be decomposed uniquely as*

$$\omega = \omega^{(2,0)} + \omega^{(1,1)} + \omega^{(0,2)},$$

*where $\omega^{(2,0)}$ and $\overline{\omega^{(0,2)}}$ are quadratic differentials compatible with the complex structure and $\omega^{(1,1)}$ is a multiple of the metric $g$.*

*Proof.* Just observe that once fixed local coordinates $u, v$ the symmetric forms $dz^2 = du^2 - dv^2 + 2idudv$, $d\overline{z} = du^2 - dv^2 - 2idudv$ and $|dz|^2 = du^2 + dv^2$ are three linearly independent elements in a three-dimensional vector space, hence a base. □

We can use this lemma to find a decomposition of the pull-back metric $(f^*(h))$. What we find is actually the quadratic differential we defined in lemma 4.17

**Lemma 4.19.** *Let $f : (\Sigma_1, g) \to (\Sigma_2, h)$ a map two Riemann surfaces with $\Sigma_1$ being hyperbolic. Let $q$ be the quadratic differential associated to $f$ as in lemma*

*4.17. Then*

$$q = (f^*(h))^{(2,0)} \ .$$

*Moreover we have that*

$$f^*(h) = q + eg + \bar{q} \ .$$

*Proof.* Let us assume as usual that $g = \lambda^2|dz|^2$ and $h = \varrho^2|dw|^2$ are the metrics respectively on $\Sigma_1$ and $\Sigma_2$, and that $w = w(z)$ is a local representation of the map $f$. Then we have

$$f^*(dw) = dw = w_z dz + w_{\overline{z}} d\overline{z} \qquad \text{and} \qquad f^*(d\overline{w}) = \overline{w}_z dz + \overline{w}_{\overline{z}} d\overline{z} \ .$$

Now, if we pull back $h$ we obtain

$$\begin{aligned} f^*(h) &= \varrho^2(w)|f^*(dw)|^2 \\ &= \varrho^2(w)(w_z dz + w_{\overline{z}} d\overline{z})(\overline{w}_z dz + \overline{w}_{\overline{z}} d\overline{z}) \\ &= \varrho^2(w)\big(w_z\overline{w}_z dz^2 + (|w_z|^2 + |w_{\overline{z}}|^2)|dz|^2 + \overline{w}_{\overline{z}} w_{\overline{z}} d\overline{z}^2\big) \ . \end{aligned}$$

Hence proving the decomposition. □

This motivates the definition of the Hopf differential

**Definition 4.20.** Given $f : (\Sigma_1, g) \to (\Sigma_2, h)$ between two Riemann surfaces, we define the Hopf differential as the quadratic differential

$$\mathrm{Hopf}(f) = (f^*(h))^{(2,0)} \ .$$

This lets us state an important result, that is a partial inverse of lemma 4.17.

**Corollary 4.21.** *If $f$ is a local diffeomorphism, so that it is orientation-preserving, then if* $\mathrm{Hopf}(f)$ *is holomorphic, $f$ is harmonic.*

*Proof.* In the proof of lemma 4.17 we showed that

$$\begin{aligned} \frac{\partial}{\partial \overline{z}}(\varrho^2(w) w_z \overline{w}_z) = \overline{w}_z\big(&\varrho^2(w) w_{z\overline{z}} + 2\varrho(w)\varrho_w(w) w_{\overline{z}} w_z\big) \\ &+ w_z \overline{\big(\varrho^2(w) w_{z\overline{z}} + 2\varrho(w)\varrho_w(w) w_{\overline{z}} w_z\big)}. \end{aligned}$$

Since $\mathrm{Hopf}(f)$ is holomorphic, the LHS is zero. Then set

$$F := \varrho^2(w) w_{z\overline{z}} + 2\varrho(w)\varrho_w(w) w_{\overline{z}} w_z.$$

$$\begin{aligned} 0 &= \frac{\partial}{\partial \overline{z}}(\varrho^2(w) w_z \overline{w_z}) \\ &\Rightarrow \overline{w_{\overline{z}}} F + w_z \overline{F} = 0 \\ &\Rightarrow \overline{w_{\overline{z}}} F = -w_z \overline{F} \qquad (4.8) \\ &\Rightarrow w_{\overline{z}} \overline{F} = -\overline{w_z} F \ . \qquad (4.9) \end{aligned}$$

By multiplying (4.8) and (4.9) we obtain

$$|w_{\overline{z}}|^2|F|^2 = |w_z|^2|F|^2,$$

hence

$$(|w_z|^2 - |w_{\overline{z}}|^2)|F|^2 = 0.$$

Since $f$ is orientation-preserving local diffeomorphism, remembering that $\overline{w_z} = \overline{w}_{\overline{z}}$,

$$|w_z|^2 - |w_{\overline{z}}|^2 = J(f) > 0,$$

hence $F = 0$ and $f$ is harmonic. □

We can calculate, pointwise, a norm for a quadratic differential. Fix $(e_i)_i$ a local orthonormal frame on $(\Sigma_1, g)$. Let us define the $g$ norm of the quadratic differential $q$ as

$$\|q\|_g = \frac{1}{4}\sum_{i,j} |q(e_i, e_j)|^2 \ .$$

It is possible to rewrite this quantity in terms of the energies of the map $f$. If the metric is $\lambda^2|dz|^2$, two orthonromal vectors are $e_1 = \frac{1}{\lambda}\frac{\partial}{\partial x}$ and $e_2 = \frac{1}{\lambda}\frac{\partial}{\partial y}$, hence

$$dz^2(e_1, e_1) = \frac{1}{\lambda^2},\ dz^2(e_1, e_2) = i\frac{1}{\lambda^2},\ dz^2(e_2, e_2) = -\frac{1}{\lambda^2} \ .$$

So that we have

$$\begin{aligned}
\|q\|_g^2 &= \frac{1}{4}\sum_{ij} |q(e_i, e_j)|^2 \\
&= \frac{1}{4}\left(|q(e_1, e_1)|^2 + 2|q(e_1, e_2)|^2 + |q(e_2, e_2)|^2\right) \\
&= \frac{1}{4}|q(z)|^2\left(|dz(e_1, e_1)|^2 + 2|q(e_1, e_2)|^2 + |q(e_2, e_2)|^2\right) \\
&= \frac{|q(z)|^2}{\lambda^4(z)} \ .
\end{aligned}$$

If we apply calculate the norm of the Hopf differential of a smooth $f : \Sigma_1 \to \Sigma_2$, with local representation $w = w(z)$,

$$\|\operatorname{Hopf}(f)\|_g^2(z) = \frac{\varrho^4(w)}{\lambda^4}|w_z|^2|w_{\overline{z}}|^2 \ .$$

### 4.5.2 The second fundamental form and the Hopf differential of the Gauss map

We now present some observations that will illustrate how the Hopf differential of the Gauss map is related to the second fundamental form.

**Lemma 4.22.** *Let $x, y$ be some coordinates. Then*

$$a(dx^2 - dy^2) + 2bdxdy = \frac{1}{2}[(a - ib)(dx + idy)^2 + (a + ib)(dx - idy)^2]$$

*Proof.* By calculation, (remember the shorthand $dxdy = \frac{1}{2}(dx \otimes dy + dy \otimes dx)$, $dx^2 = dx \otimes dx$, $dy^2 = dy \otimes dy$)

$$\begin{aligned}(a - ib)(dx + idy)^2 &= (a - ib)(dx^2 - dy^2 + 2idxdy)\\ (a + ib)(dx - idy)^2 &= (a + ib)(dx^2 - dy^2 - 2idxdy)\ ,\end{aligned}$$

so that

$$\begin{aligned}&(a - ib)(dx + idy)^2 + (a + ib)(dx - idy)^2 =\\ &\quad = adx^2 - ady^2 + 2aidxdy - ibdx^2 + ibdy^2 + 2bdxdy\\ &\quad + adx^2 - ady^2 - 2aidxdy + ibdx^2 - ibdy^2 + 2bdxdy =\\ &\quad = 2a(dx^2 - dy^2) + 4bdxdy\ .\end{aligned}$$

□

Hence if we set $z = x + iy, \bar{z} = x - iy$ we obtain

$$a(dx^2 - dy^2) + 2bdxdy = \frac{1}{2}\left[(a - ib)dz^2 + (a + ib)d\bar{z}^2\right]\ .$$

Let us now take $\alpha$ a symmetric 2-form on $S$. Then there is a morphism of the tangent bundle $L : TM \to TM$ such that

$$\alpha(\cdot, \cdot) = \mathbf{I}(L\cdot, \cdot)\ .$$

Which is self-adjoint, since $\alpha$ is symmetric by hypothesis. $L$ is at every point a linear endomorphism, and we can decompose it using $\mathbf{I}$ in a traceless part and a diagonal part,

$$L = \frac{\operatorname{tr} L}{2}\operatorname{Id} + L_0\ .$$

Now, let us suppose that $(x, y)$ are isothermal coordinates. We can write $L_0$ as

$$\begin{aligned}L_0(\partial_x) &= a\partial_x + b\partial_y\\ L_0(\partial_y) &= c\partial_x + d\partial_y\ ,\end{aligned}$$

and since $L_0$ is symmetric and traceless, we have $d = -a$ and $c = b$. Hence we can represent $L_0$ with the matrix

$$[L_0] = \begin{bmatrix} a & b\\ b & -a\end{bmatrix}\ .$$

Now, as in lemma 4.18 we may suppose to write $\alpha$ as

$$\alpha = \alpha^{(2,0)} + \alpha^{(1,1)} + \alpha^{(0,2)}\ . \tag{4.10}$$

We may observe that, in the decomposition (4.10), from the theorem we have that $\alpha^{(2,0)}$ and $\alpha^{(0,2)}$ are locally multiples of $dz^2$ and $d\bar{z}^2$ respectively, which are traceless. Hence we observe

$$\begin{aligned}\alpha = \alpha^{(2,0)} + \alpha^{(1,1)} + \alpha^{(0,2)} &= \mathbf{I}(L\cdot,\cdot) \\ &= \mathbf{I}\left(\left(\tfrac{1}{2}\operatorname{tr} L\,\mathrm{Id} + L_0\right)\cdot,\cdot\right) \\ &= \tfrac{1}{2}\operatorname{tr}(L)\mathbf{I}(\cdot,\cdot) + \mathbf{I}(L_0\cdot,\cdot)\ .\end{aligned}$$

By separating the traceless part and the trace part we have

$$\alpha^{(1,1)} = \tfrac{1}{2}\operatorname{tr}(L)\mathbf{I}, \qquad \alpha^{(2,0)} + \alpha^{(0,2)} = \mathbf{I}(L_0\cdot,\cdot)\ . \tag{4.11}$$

Also, again from 4.18, since $\alpha$ is real valued, we have $\alpha^{(0,2)} = \overline{\alpha^{(2,0)}}$.
We have proved the following lemma

**Lemma 4.23.** *$\alpha$ symmetric 2-form on $S$ with values in $\mathbb{R}$. Then $\alpha$ is traceless if and only if there is a quadratic differential $\alpha^{(2,0)} = \Phi$ such that*

$$\alpha = \tfrac{1}{2}\left(\Phi + \overline{\Phi}\right)\ .$$

We may now prove the following important lemma

**Lemma 4.24.** *With the notation above, it holds that*

$$\alpha^{(2,0)} = \tfrac{1}{2}(a - ib)dz^2\ .$$

*Proof.* Let us fix $v = v_x\partial_x + v_y\partial_y$, by looking at the associated quadratic form to $\mathbf{I}(L_0\cdot,\cdot)$,

$$\begin{aligned}\mathbf{I}(L_0(v_x\partial_x + v_y\partial_y), v_x\partial x + v_y\partial_y) &= \mathbf{I}(v_xL_0\partial_x + v_yL_0\partial_y, v_x\partial x + v_y\partial_y) \\ &= \mathbf{I}(v_x(a\partial_x + b\partial_y) + v_y(b\partial_x - a\partial_y), v_x\partial x + v_y\partial_y) \\ &= e^{2u}\left(av_x^2 + bv_xv_y + bv_xv_y - av_y^2\right)\ ,\end{aligned}$$

we have

$$\begin{aligned}\mathbf{I}(L_0\cdot,\cdot) &= a(dx^2 - dy^2) + 2bdxdy \\ \text{[Lemma 4.22]}\quad &= \tfrac{1}{2}(a - ib)(dx + idy)^2 + \tfrac{1}{2}(a + ib)(dx - idy)^2\ .\end{aligned}$$

Therefore

$$\alpha^{(2,0)} = \tfrac{1}{2}(a - ib)dz^2\ , \tag{4.12}$$

as we intended. □

Let now $\alpha$ be the second fundamental form $\mathbf{II}$, so that $L$ is the shape operator $A$. We consider the Gauss map $N : \mathbb{D} \to \mathbb{H}$ and the pull-back of the metric of

$\mathbb{H}$ through $N$, which we can write as

$$\begin{aligned} N^*(g_{\mathbb{H}^2})(v,w) &= \langle D_v N, D_w N\rangle \\ \text{[def. of } A\text{]} \qquad &= \mathbf{I}(A(v), A(w)) \\ [A \text{ is self-adjoint}] \qquad &= \mathbf{I}(A^2 v, w) \ . \end{aligned} \tag{4.13}$$

We can apply Cayley-Hamilton's theorem on $p_A(x) = x^2 - \mathrm{tr}(A)x + \det(A)$, obtaining

$$p_A(A) = 0 \Rightarrow A^2 = \mathrm{tr}(A)A - \det(A)\mathbb{1} \ .$$

Recalling that $\mathrm{tr}(A) = 2H$ and $\det(A) = -K$, we obtain

$$\begin{aligned} A^2 &= 2HA + K\,\mathrm{Id} \\ &= 2H(A_0 + H\,\mathrm{Id}) + K\,\mathrm{Id} \\ &= 2HA_0 + (2H^2 + K)\,\mathrm{Id} \ . \end{aligned} \tag{4.14}$$

Which implies in particular that $e = 2H^2 + K$. Now, the relations (4.13) and (4.14) along with the general (4.11) give us

$$N^*(g_{\mathbb{H}})^{(2,0)} + \overline{N^*(g_{\mathbb{H}})^{(2,0)}} = 2H\,\mathbf{I}(A_0\cdot,\cdot) \ .$$

So that by applying the identity (4.12) to $\alpha = 2H\,\mathbf{II}_0$ we obtain

$$N^*(g_{\mathbb{H}})^{(2,0)} = H(a - ib)dz^2 \ .$$

And we can rewrite

$$\mathrm{Hopf}(N) + \overline{\mathrm{Hopf}(N)} = 2H\,\mathbf{I}(A_0\cdot,\cdot) \ .$$

As usual let us write $\mathbf{II}$ locally as

$$\mathbf{II} = h_{ij}dx^i dx^j, \quad \text{and suppose} \quad \mathrm{Hopf}(N) = \varphi\, dz^2.$$

Since $H = 1$ and $\mathbf{I} = e^{2w}(dx^2 + dy^2)$, we have

$$h_{11} + h_{22} = 2e^{2w}.$$

Moreover, from

$$2\mathbf{II}_0 = \mathrm{Hopf}(N) + \overline{\mathrm{Hopf}(N)} = \varphi\, dz^2 + \overline{\varphi}\, d\bar{z}^2$$

and Lemma 4.22, writing $\varphi = a - ib$, we obtain

$$\begin{aligned} 2\mathbf{II}_0 &= 2a(dx^2 - dy^2) + 4b\,dx\,dy \\ &= (h_{11} - h_{22})(dx^2 - dy^2) + 4h_{12}dx\,dy, \end{aligned}$$

hence

$$h_{11} - h_{22} = 2\Re(\varphi), \qquad h_{12} = -\Im(\varphi).$$

We note that these relations can be used to reconstruct the coefficients $(h_{ij})$, and therefore the second fundamental form $\mathbf{II}$, from the data $w$ and $\varphi$.

$$\begin{cases} h_{11} + h_{22} = 2e^{2w} \\ h_{11} - h_{22} = 2\Re(\varphi) \\ h_{12} = -\Im(\varphi) \end{cases} . \tag{4.15}$$

## 4.6 A different form for Gauss' equation

We shall now prove the equivalence of Gauss' equation with a certain PDE on $\mathbb{D}$, which we will derive with respect to the Euclidean backgroun metric $|dz|^2$ and then rewrite it with respect to the hyperbolic metric.

Let $\mathbf{I} = e^{2w}|dz|^2$, and let $(x, y)$ be the corresponding isothermal coordinates, with associated complex coordinate $z = x + iy$. It is clear that

$$\det(g_{ij}) = e^{4w} \quad .$$

Let moreover

$$\mathbf{II} = h_{ij}\, dx^i dx^j \quad \text{and} \quad \Phi = \mathrm{Hopf}(N) = \varphi\, dz^2,$$

where $N$ is the Gauss map of the immersion $S \hookrightarrow \mathbb{M}^{2,1}$. Recall Gauss' equation:

$$K = -\det_{\mathbf{I}}(\mathbf{II}) = -\frac{\det(h_{ij})}{\det(g_{ij})} \quad .$$

We already know the expression of $\det(g_{ij})$, we need to rewrite $\det(h_{ij})$.

$$\begin{aligned} \det(h_{ij}) &= h_{11}h_{22} - h_{12}^2 \\ &= \tfrac{1}{4}\left[(h_{11} + h_{22})^2 - (h_{11} - h_{22})^2\right] - h_{12}^2 \\ \text{[(4.15)]} \quad &= \tfrac{1}{4}\left[(2e^{2w})^2 - (2\Re(\varphi))^2\right] - \Im(\varphi)^2 \\ &= e^{4w} - \Re(\varphi)^2 - \Im(\varphi)^2 \\ &= e^{4w} - |\varphi|^2 \quad . \end{aligned}$$

Now we recall that

$$K = -e^{-2w}\Delta w \quad .$$

So, substituting in Gauss' equation, we obtain

$$\begin{aligned} -e^{-2w}\Delta w &= -\frac{e^{4w}-|\varphi|^2}{e^{4w}} \\ &= -1+|\varphi|^2 e^{-4w} \quad , \end{aligned}$$

and therefore

$$\Delta w = e^{2w} - |\varphi|^2 e^{-2w} \quad . \tag{4.16}$$

This is the Euclidean form of the equation, which will be studied in the later sections.

For completeness, let us now rewrite the same equation with respect to the hyperbolic background metric

$$ds_p^2 = \frac{4}{(1-|z|^2)^2}|dz|^2.$$

If the first fundamental form is expressed by

$$\mathbf{I} = e^{2\tilde{w}} ds_p^2,$$

then

$$e^{2w}|dz|^2 = e^{2\tilde{w}}\frac{4}{(1-|z|^2)^2}|dz|^2,$$

hence

$$w = \tilde{w} + \log\left(\frac{2}{1-|z|^2}\right) \quad . \tag{4.17}$$

Moreover,

$$\Delta_p = \frac{(1-|z|^2)^2}{4}\Delta$$

and

$$\Delta \log\left(\frac{2}{1-|z|^2}\right) = \frac{4}{(1-|z|^2)^2}.$$

Therefore

$$\begin{aligned} \Delta w &= \Delta\tilde{w} + \Delta\log\left(\frac{2}{1-|z|^2}\right) \\ &= \frac{4}{(1-|z|^2)^2}\Delta_p\tilde{w} + \frac{4}{(1-|z|^2)^2} \\ &= \frac{4}{(1-|z|^2)^2}\left(\Delta_p\tilde{w}+1\right) \quad . \end{aligned}$$

On the other hand, by (4.17),

$$e^{2w} = \frac{4}{(1-|z|^2)^2}e^{2\tilde{w}}.$$

Also, if $\|\Phi\|$ denotes the norm of $\Phi$ with respect to the Poincaré metric $ds_p^2$, then

$$\|\Phi\|^2 = \frac{(1-|z|^2)^4}{4^2}|\varphi|^2,$$

that is,

$$|\varphi|^2 = \frac{16}{(1-|z|^2)^4}\|\Phi\|^2.$$

Hence

$$|\varphi|^2 e^{-2w} = \frac{4}{(1-|z|^2)^2}\|\Phi\|^2 e^{-2\tilde{w}}.$$

Substituting these expressions into (4.16), we get

$$\frac{4}{(1-|z|^2)^2}\left(\Delta_p \tilde{w} + 1\right) = \frac{4}{(1-|z|^2)^2}e^{2\tilde{w}} - \frac{4}{(1-|z|^2)^2}\|\Phi\|^2 e^{-2\tilde{w}} \quad ,$$

and therefore

$$\Delta_p \tilde{w} = e^{2\tilde{w}} - \|\Phi\|^2 e^{-2\tilde{w}} - 1 \quad . \tag{4.18}$$

This is exactly the form of the PDE found in [10]. Again, it might be useful to summarize this result in a theorem statement.

**Theorem 4.25.** *Let $S$ be a constant mean curvature 1 surface, let $N : \mathbb{D} \to \mathbb{H}$ be its Gauss map, and let $\Phi = \phi\, dz^2$ be its Hopf differential. If the induced metric on $S$ is written as*

$$\mathbf{I} = e^{2w} ds_p^2,$$

*then the Gauss equation for $S$ is equivalent to*

$$\Delta_p w = e^{2w} - \|\Phi\|^2 e^{-2w} - 1.$$

# Chapter 5

# CMC surfaces and harmonic maps

We now introduce the class of surfaces that will be studied throughout this section.

**Definition 5.1.** A constant mean curvature cut is a complete entire spacelike surface $S \subset \mathbb{M}^{2,1}$ with constant mean curvature $H > 0$.

This is essentially the same setting we exposed in Section 2 if not for the additional condition of having a constant mean curvature. The first step for classification is the following proposition.

**Proposition 5.2.** *Let $S$ be a complete spacelike surface in $\mathbb{M}^{2,1}$ with constant mean curvature $H$. $S$ can be represented as the graph of an entire convex function $x^3 = f(x^1, x^2)$. The Gaussian curvature of $S$ satisfies $K \leq 0$. More precisely, either $K < 0$ everywhere or if $K = 0$ at some point then $K \equiv 0$ and the surface is parabolic.*

An immediate consequence of being the graph of a convex function is that the second fundamental form of the surface is positive definite, from which we get that the mean curvature must be positive.

We observe now that we may suppose the mean curvature of $S$ to be 1. In fact, if $\iota : S \hookrightarrow \mathbb{M}^{2,1}$ is an immersion with constant mean curvature $H = k > 0$, then we may consider the dilated immersion

$$\tilde{\iota} = \mu\iota,$$

which has constant mean curvature $\tilde{H} = \mu^{-1}k$. Choosing $\mu = k$, we obtain a surface with constant mean curvature 1.

Therefore, there exists a bijection between complete space-like surfaces with constant mean curvature $k$ and complete space-like surfaces with constant mean curvature 1. Hence a classification of the latter immediately yields a classification of the former. We may then assume from now on that the surfaces have constant mean curvature 1.

Let us then fix $S$ a hyperbolic CMC cut with mean curvature 1. The hyperbolic hypothesis means that using the uniformization theorem we can find a conformal diffeomorphism

$$\sigma : \mathbb{D} \to S.$$

Let us now consider the Gauss map of $S$, $u : S \to \mathbb{D}$. By composing with $\sigma$ read it as a map

$$u : \mathbb{D} \to \mathbb{D}.$$

The hypothesis on the mean curvature of $S$ being constant lets us apply the Theorem 4.15 and obtain that $u$ is harmonic (we remind that $\mathbb{D}$ is isometric to $\mathbb{H}$ found in the statement of the theorem).

Now, we can pull back the metric of $S$ through the conformal parametrization and express it in terms of $u$. We provide its proof at the endo of the section.

**Proposition 5.3.** *Suppose that $S$ has constant mean curvature 1. Then the pulled-back induced metric $g = \sigma^*(\mathbf{I})$ of $S$ and the respective Gaussian curvature are given by*

$$g = \frac{4}{(1-|u|^2)^2}|u_z|^2|dz|^2, \qquad K = \frac{|u_{\overline{z}}|^2}{|u_z|^2} - 1 \quad .$$

*Proof.* Let us denote by $N : S \to \mathbb{H}$ the Gauss map with values in the hyperboloid, and by $j$ the isometry $j : \mathbb{H} \to \mathbb{D}$. Then we use the notation

$$u = j \circ N : S \to \mathbb{D}.$$

As usual, we fix isothermal coordinates $(x, y)$ on $S$ and set $z = x + iy$. Denote by $\sigma : \mathbb{D} \to S$ the corresponding local parametrization. Then we express the metric of $S$ in these coordinates as

$$g = \sigma^*(\mathbf{I}) = e^{2w}|dz|^2 \ . \tag{5.1}$$

By abuse of notation, we still denote by $u$ and $N$ the local expressions $u \circ \sigma$ and $N \circ \sigma$ on $\mathbb{D}$.

Then

$$\langle \sigma_x, \sigma_x \rangle = \langle \sigma_y, \sigma_y \rangle = e^{2w}, \qquad \langle \sigma_x, \sigma_y \rangle = 0. \tag{5.2}$$

Let $A$ be the shape operator of the surface $S$. Since the mean curvature is 1, we have

$$\mathrm{tr}(A) = 2.$$

Moreover $A$ is self-adjoint with respect to the induced metric, hence, in the

orthogonal basis $(\sigma_x, \sigma_y)$, it can be represented by

$$[A] = \begin{pmatrix} 1+a & b \\ b & 1-a \end{pmatrix}, \qquad \text{equivalently} \qquad \begin{cases} A(\sigma_x) = (1+a)\sigma_x + b\sigma_y, \\ A(\sigma_y) = b\sigma_x + (1-a)\sigma_y, \end{cases} \tag{5.3}$$

for some real-valued functions $a, b$.

Now let $J$ be the positive rotation of angle $\pi/2$ on the tangent plane, so that

$$J\sigma_x = \sigma_y, \qquad J\sigma_y = -\sigma_x. \tag{5.4}$$

Using the chain rule, and recalling that $A = DN$, we have

$$\begin{aligned} u_x &= dj_N(DN(\sigma_x)) & u_y &= dj_N(DN(\sigma_y)) \\ &= dj_N(A(\sigma_x)), & &= dj_N(A(\sigma_y)). \end{aligned} \tag{5.5}$$

Using that $j$ is an isometry, we obtain

$$|u_x|_h^2 = |A(\sigma_x)|_{g_{\mathbb{H}}}^2, \qquad |u_y|_h^2 = |A(\sigma_y)|_{g_{\mathbb{H}}}^2.$$

Since the hyperbolic metric on $\mathbb{D}$ is given by

$$h = \lambda(u)^2 |du|^2, \qquad \lambda(u) = \frac{2}{1-|u|^2}, \tag{5.6}$$

this becomes

$$\lambda(u)^2 |u_x|^2 = |A(\sigma_x)|_{g_{\mathbb{H}}}^2, \qquad \lambda(u)^2 |u_y|^2 = |A(\sigma_y)|_{g_{\mathbb{H}}}^2. \tag{5.7}$$

In the following, we omit the subscript $g_{\mathbb{H}}$ from the norms and scalar products on $T\mathbb{H}$.

Also, we observe that using the coordinates $z = x + iy$, the multiplication by $i$ on $\mathbb{D}$ corresponds to a positive rotation $J$ on the tangent plane, so that, recalling that $u_z = \frac{1}{2}(u_x - iu_y)$ and $u_{\bar{z}} = \frac{1}{2}(u_x + iu_y)$,

$$iu_y = dj_N(JA(\sigma_y)) \ .$$

In particular, recalling that $|dz|^2 = \Re(dz \otimes d\bar{z})$ and using that $j$ is an isometry, we have

$$\lambda^2(u)\Re(iu_y\overline{u_x}) = h(iu_y, u_x) = \langle JA(\sigma_y), A(\sigma_x)\rangle \ . \tag{5.8}$$

Then we obtain

$$\begin{aligned} 4\lambda^2(u)|u_z|^2 &= \lambda^2(u)|u_x - iu_y|^2 \\ &= \lambda^2(u)(u_x - iu_y)\overline{(u_x - iu_y)} \\ &= \lambda^2(u)(u_x - iu_y)(\overline{u}_x + i\overline{u}_y) \\ &= \lambda^2(u)(u_x\overline{u}_x + u_y\overline{u}_y + iu_x\overline{u}_y - iu_y\overline{u}_x) \\ \text{[(5.8)]} \quad &= \lambda^2(u)\big(|u_x|^2 + |u_y|^2 - 2\Re(iu_y\overline{u_x})\big) \\ \text{[(5.7)]} \quad &= |A(\sigma_x)|^2 + |A(\sigma_y)|^2 - 2\langle JA(\sigma_y), A(\sigma_x)\rangle. \end{aligned} \tag{5.9}$$

Moreover, using (5.3) and (5.2),

$$\begin{aligned}
|A(\sigma_x)|^2 &= \langle (1+a)\sigma_x + b\sigma_y, (1+a)\sigma_x + b\sigma_y\rangle \\
&= e^{2w}((1+a)^2 + b^2), && (5.10)\\
|A(\sigma_y)|^2 &= \langle b\sigma_x + (1-a)\sigma_y, b\sigma_x + (1-a)\sigma_y\rangle \\
&= e^{2w}((1-a)^2 + b^2), && (5.11)\\
\langle JA(\sigma_y), A(\sigma_x)\rangle &= \langle b\sigma_y - (1-a)\sigma_x, (1+a)\sigma_x + b\sigma_y\rangle \\
&= e^{2w}(a^2 + b^2 - 1). && (5.12)
\end{aligned}$$

Substituting (5.10), (5.11), and (5.12) into (5.9) we obtain

$$4\lambda(u)^2|u_z|^2 = 4e^{2w}, \quad \Rightarrow \quad \lambda(u)^2|u_z|^2 = e^{2w}. \tag{5.13}$$

Since $g = e^{2w}|dz|^2$, we conclude that

$$g = \frac{4}{(1-|u|^2)^2}|u_z|^2|dz|^2.$$

For the curvature formula, we compute similarly

$$\begin{aligned}
4e^{2w} \overset{[(5.13)]}{=} 4\lambda(u)^2|u_{\bar z}|^2 &= \lambda(u)^2(|u_x|^2 + |u_y|^2 + 2\Re(iu_y\overline{u_x})) \\
&= |A(\sigma_x)|^2 + |A(\sigma_y)|^2 + 2\langle JA(\sigma_y), A(\sigma_x)\rangle \\
&= 4(a^2+b^2)e^{2w}.
\end{aligned}$$

Therefore

$$\frac{|u_{\bar z}|^2}{|u_z|^2} = a^2 + b^2 \ . \tag{5.14}$$

On the other hand, we have that

$$\det(A) = \det([A]) = 1 - a^2 - b^2 \ ,$$

Then we recall that, for Gauss' equation,

$$K = -\det A = a^2 + b^2 - 1.$$

Substituting 5.14 in the last equation we obtain

$$K = \frac{|u_{\bar z}|^2}{|u_z|^2} - 1.$$

Which ends the proof. □

We will also need the following lemma.

**Lemma 5.4.** *The Gauss map $u$ satisfies*

$$J(u) > 0 \qquad \textit{and} \qquad \mathcal{H}(u) > 0 \ .$$

*Proof of Claim 5.4.* First, by Proposition 5.3 we have that the metric $g := \Lambda^*(\mathbf{I}_S)$ is expressed by

$$g = \frac{4}{(1-|u|^2)^2}|u_z|^2|dz|^2 \ .$$

Then we recall that the metric on $\mathbb{D}$ at a point $v$ is $ds_p^2|_v := \frac{4}{(1-|v|^2)^2}|dz|^2$. So we have

$$\begin{aligned} g &= \frac{4}{(1-|u|^2)^2}|u_z|^2|dz|^2 \\ &= ds_p^2|_u|u_z|^2 \\ &= ds_p^2|_u\mathcal{H}(u) \ \ . \end{aligned}$$

We know that both $g$ and $ds_p^2$ are Riemannian metrics, so that we obtain $\mathcal{H}(u) > 0$.

Now, from the second equation of Proposition 5.3 we have

$$K = \frac{|u_{\overline{z}}|^2}{|u_z|^2} - 1$$

so that

$$\begin{aligned} |u_z|^2K &= |u_{\overline{z}}|^2 - |u_z|^2 \\ &= -J(u) \ \ . \end{aligned}$$

Then

$$J(u) = -|u_z|^2K > 0 \ ,$$

since $K < 0$, which is true by applying Proposition 5.2 and recalling that $S$ is hyperbolic. □

Now we observe that, since we know that $S$ has constant mean curvature, we know from Theorem 4.15 that $u$ is harmonic. Then by applying Lemma 4.17 we obtain that $\mathrm{Hopf}(u)$ is holomorphic. Furthermore we observe that from the proof of the last lemma, the metric $g$ of $S$ is expressed on $\mathbb{D}$ by $e^{2w}ds_p^2$, where $w = \frac{1}{2}\log\mathcal{H}(u)$, which is well defined since $\mathcal{H} > 0$.

We have thus proved the following theorem:

**Theorem 5.5.** *Let $S$ be a hyperbolic constant mean curvature cut with mean curvature equal to 1, let $u : \mathbb{D} \to \mathbb{D}$ be the Gauss map of $S$ read through the conformal parametrization $\sigma$ and $\Phi = \varphi(z)dz^2$ its Hopf differential. Then*

- *$\Phi$ is holomorphic as a quadratic differential,*
- *$u$ preserves the orientation, so that $\mathcal{J}(u) > 0$,*
- *$w := \frac{1}{2}\log\mathcal{H}(u)$ is a well defined function on $\mathbb{D}$,*
- *$e^{2w}ds_p^2$ is the metric of $S$ read through the conformal parametrization $\sigma$,*

# Chapter 6

# Classification of hyperbolic CMC cuts

We will need first to make some observations on quadratic differentials, that we will use in the proof of the classification theorem.

### 6.0.1 The codazzi equation and holomorphic quadratic differentials

Let $g = e^{2w}(dx^2 + dy^2)$ be a conformal metric on $\mathbb{D}$, and let $h$ be a symmetric real 2-form on $(\mathbb{D}, g)$.

By applying Lemma 4.23 we obtain that $h = \Phi + \overline{\Phi}$ for a certain quadratic differential $\Phi$. We may now prove that, if $h$ satisfies the Codazzi equations if and only if $\Phi$ is holomorphic.

Let us choose local isothermal coordinates $(x, y)$. Writing locally

$$\Phi = \tfrac{1}{2}(a - ib)dz^2 \ ,$$

we have

$$h = \Phi + \bar{\Phi} = a(dx^2 - dy^2) + 2b\, dxdy \ ,$$

so that, with respect to the frame $(\partial_x, \partial_y)$, $h$ is represented by

$$[h] = \begin{pmatrix} a & b \\ b & -a \end{pmatrix} \ .$$

In particular, $h$ is symmetric and traceless with respect to $g$.

We now recall the definition of the derivative of a tensor,

$$(\nabla_X h)(Y, Z) = X(h(Y, Z)) - h(\nabla_X Y, Z) - h(Y, \nabla_X Z) \ ,$$

so that we may explicit the equation in the basis. In order to do that we need

the Christoffel symbols for the metric $g = e^{2w}(dx^2 + dy^2)$.
Let $\Gamma_{ij}^k$ be such that

$$\nabla_{\partial_i}\partial_j = \Gamma_{ij}^k \partial_k \ .$$

Since $\langle \partial_x, \partial_x \rangle = \langle \partial_y, \partial_y \rangle = e^{2w}$ and $\langle \partial_x, \partial_y \rangle = 0$, the compatibility with the metric yields

$$\begin{aligned}
\Gamma_{11}^1 e^{2w} &= \langle \nabla_{\partial_x}\partial_x, \partial_x \rangle = \frac{1}{2}\partial_x \langle \partial_x, \partial_x \rangle = e^{2w} w_x \\
\Gamma_{11}^2 e^{2w} &= \langle \nabla_{\partial_x}\partial_x, \partial_y \rangle = -\frac{1}{2}\partial_y \langle \partial_x, \partial_x \rangle = -e^{2w} w_y,
\end{aligned}$$

hence $\Gamma_{11}^1 = w_x$ and $\Gamma_{11}^2 = -w_y$. Similarly,

$$\begin{aligned}
\Gamma_{12}^1 e^{2w} &= \langle \nabla_{\partial_x}\partial_y, \partial_x \rangle = \frac{1}{2}\partial_y \langle \partial_x, \partial_x \rangle = e^{2w} w_y, \\
\Gamma_{12}^2 e^{2w} &= \langle \nabla_{\partial_x}\partial_y, \partial_y \rangle = \frac{1}{2}\partial_x \langle \partial_y, \partial_y \rangle = e^{2w} w_x.
\end{aligned}$$

so $\Gamma_{12}^1 = w_y$ and $\Gamma_{12}^2 = w_x$. Since the connection is torsion free, $\Gamma_{21}^k = \Gamma_{12}^k$. Finally,

$$\begin{aligned}
\Gamma_{22}^1 e^{2w} &= \langle \nabla_{\partial_y}\partial_y, \partial_x \rangle = -\frac{1}{2}\partial_x \langle \partial_y, \partial_y \rangle = -e^{2w} w_x, \\
\Gamma_{22}^2 e^{2w} &= \langle \nabla_{\partial_y}\partial_y, \partial_y \rangle = \frac{1}{2}\partial_y \langle \partial_y, \partial_y \rangle = e^{2w} w_y.
\end{aligned}$$

so $\Gamma_{22}^1 = -w_x$ and $\Gamma_{22}^2 = w_y$. Therefore

$$\begin{aligned}
\nabla_{\partial_x}\partial_x &= w_x \partial_x - w_y \partial_y \ , \\
\nabla_{\partial_x}\partial_y &= w_y \partial_x + w_x \partial_y = \nabla_{\partial_y}\partial_x \ , \\
\nabla_{\partial_y}\partial_y &= -w_x \partial_x + w_y \partial_y \ .
\end{aligned}$$

We observe that $h$ is symmetric, and therefore $\nabla h$ is symmetric in the last two entries, therefore the Codazzi equations in the basis $(\partial_x, \partial_y)$ reduce to

$$\begin{cases}
(\nabla_{\partial_x} h)(\partial_x, \partial_y) &= (\nabla_{\partial_y} h)(\partial_x, \partial_x), \\
(\nabla_{\partial_x} h)(\partial_y, \partial_y) &= (\nabla_{\partial_y} h)(\partial_x, \partial_y).
\end{cases}$$

We can calculate $(\nabla_{\partial_x} h)(\partial_x, \partial_y)$ and $(\nabla_{\partial_y} h)(\partial_x, \partial_x)$,

$$\begin{aligned}
(\nabla_{\partial_x} h)(\partial_y, \partial_x) &= \partial_x\big(h(\partial_y, \partial_x)\big) - h(\nabla_{\partial_x}\partial_y, \partial_x) - h(\partial_y, \nabla_{\partial_x}\partial_x) \\
&= \partial_x b - h(w_y\partial_x + w_x\partial_y, \partial_x) - h(\partial_y, w_x\partial_x - w_y\partial_y) \\
&= b_x - (w_y a + w_x b) - (w_x b + w_y a) \\
&= b_x - 2w_x b - 2w_y a, \\
(\nabla_{\partial_y} h)(\partial_x, \partial_x) &= \partial_y\big(h(\partial_x, \partial_x)\big) - h(\nabla_{\partial_y}\partial_x, \partial_x) - h(\partial_x, \nabla_{\partial_y}\partial_x) \\
&= \partial_y a - h(w_y\partial_x + w_x\partial_y, \partial_x) - h(\partial_x, w_y\partial_x + w_x\partial_y) \\
&= a_y - (w_y a + w_x b) - (w_y a + w_x b) \\
&= a_y - 2w_y a - 2w_x b.
\end{aligned}$$

Hence

$$(\nabla_{\partial_x} h)(\partial_x, \partial_y) = (\nabla_{\partial_y} h)(\partial_x, \partial_x) \Leftrightarrow b_x = a_y \ .$$

Developing the calculations in the same way we obtain the equivalence

$$(\nabla_{\partial_x} h)(\partial_y, \partial_y) = (\nabla_{\partial_y} h)(\partial_x, \partial_y) \Leftrightarrow b_y = -a_x \ .$$

So that the Codazzi equations reduce to

$$\begin{cases} a_x = (-b)_y \\ a_y = -(-b)_x \end{cases} ,$$

which are the Cauchy-Riemann equations for $\varphi = a - ib$. We can summarize the content of this section in the following statement.

**Proposition 6.1.** *Let $g = e^{2\tilde{w}}|dz|^2$ be a conformal metric on $\mathbb{D}$, and let $\Phi$ be a quadratic differential. If $h = \Phi + \bar{\Phi}$, then $h$ is a symmetric traceless bilinear form. Moreover,*

$$h \text{ satisfies the Codazzi equations for } g \quad \Leftrightarrow \quad \Phi \text{ is holomorphic} \ .$$

## 6.1 The classification theorem

In Section 5 we essentially followed the sequence of constructions

$$S \longmapsto u \longmapsto (\Phi, w) \longmapsto \Phi, \tag{6.1}$$

where $S$ is a hyperbolic CMC-1 cut in $\mathbb{M}^{2,1}$; $u : \mathbb{D} \to \mathbb{D}$ is the Gauss map of $S$ read through the conformal parametrization obtained by the uniformization theorem; $\Phi$ is the Hopf differential of $u$ and $w = \frac{1}{2}\log\mathcal{H}(u)$ with $\mathcal{H}$ denoting the holomorphic energy of $u$. Furthermore we showed that the involved quantities satisfy the properties in Theorem 5.5

As it is, the assignment from the surface $S$ to the holomorphic quadratic differential on $\mathbb{D}$, $S \mapsto \Phi$, is not well defined, because it depends on the choice

of the conformal parametrization. This ambiguity disappears after passing to equivalent classes by considering the map

$$\mathscr{A} : \mathbf{E} \to \mathbf{Q}_{\mathbb{D}} \quad ,$$

where $\mathbf{E}$ is the space of surfaces $S$ up to isometry and $\mathbf{Q}_{\mathbb{D}}$ is the space of holomorphic quadratic differentials up to change of conformal coordinates. This is in fact justified because we can't distinguish from an intrinsic point of view two isometric surfaces, while a quadratic differential compatible with the complex structure should be intended independently from its local representation.
So we can prove the theorem:

**Theorem 6.2.** *The map $\mathscr{A}$ is well defined and bijective.*

The proof of this theorem requires an existence and uniqueness result for the conformal factor of the metric induced on $S$, expressed in terms of a holomorphic quadratic differential which will later be identified with the Hopf differential of the Gauss map. We state this result here and provide proof to the next sections.

**Claim 6.3.** *Let $\Phi$ be a holomorphic quadratic differential on $\mathbb{D}$. Then there exists a unique smooth function $w : \mathbb{D} \to \mathbb{R}$ such that*

$$\Delta_p w = e^{2w} - \|\Phi\|^2 e^{-2w} - 1,$$

*the metric $e^{2w} ds_p^2$ is complete and $e^{2w} - \|\Phi\|^2 e^{-2w} > 0$.*

*Proof of Theorem 6.2.*

**1.** Well-definedness:
We just observe that by choosing a different $S' \in [S]$ ultimately results to the same map $u$ post-composed with an isometry of the hyperbolic disk $\mathbb{D}$. The Hopf differential associated to $u$ remains unchanged under this transformation.

**2.** Surjectivity:
Let $[\Phi] \in \mathbf{Q}_{\mathbb{D}}$ and choose a representative

$$\Phi = \varphi(z)\, dz^2.$$

We reconstruct the surface $S$ starting from $\Phi$ using the following diagram

$$\begin{tikzcd}
\Phi \arrow[r, mapsto, "(*1)"] & \|\Phi\| \arrow[r, mapsto, "(*2)"] & w \arrow[r, mapsto, "(*5)"] \arrow[d, mapsto, "(*3)"] & (h_{ij})_{ij} \arrow[dr, mapsto] & & \\
& & & & \text{G.C.} \arrow[r, "(*6)"] & S \\
& & e^{2w} ds_p^2 \arrow[r, mapsto, "(*4)"] & (g_{ij})_{ij} \arrow[ur, mapsto] & &
\end{tikzcd}$$

.

(∗1) Starting from the holomorphic quadratic differential $\Phi$, we compute its norm $\|\Phi\|$ with respect to the Poincaré metric.
(∗2) We recover the function $w$ using Claim 6.3.
(∗3) We define the bilinear form

$$\mathbf{I} = e^{2\tilde{w}} |dz|^2 = e^{2w} ds_p^2 \ .$$

Which, by the choice of $w$, it defines a complete metric on $\mathbb{D}$.

(∗4) If the first fundamental form is

$$\mathbf{I} = e^{2\tilde{w}}(dx^2 + dy^2),$$

then its coefficients in the coordinates $(x, y)$ are

$$(g_{ij})_{ij} = e^{2\tilde{w}} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} .$$

(∗5) We now define a candidate second fundamental form

$$\mathbf{II} = h_{ij} dx^i dx^j$$

by requiring that the relation we found in (4.15) hold,

$$\begin{cases} \frac{1}{2}(h_{11} + h_{22}) = e^{2\tilde{w}} \\ h_{11} - h_{22} = 2\Re(\varphi) \\ h_{12} = -\Im(\varphi) \ . \end{cases} \tag{6.2}$$

This is a linear system which can be easily solved to find $(h_{ij})_{ij}$.

(∗6) We claim that the pair $(\mathbf{I}, \mathbf{II})$ satisfies the Gauss-Codazzi equations.

First we observe that, in virtue of Equation (6.2), we have that the symmetric real bilinear $\mathbf{II}$ form satisfies

$$\mathbf{II} = \Phi + e^{2\tilde{w}}|dz|^2 + \overline{\Phi} \ .$$

Then when we look at the traceless part $\mathbf{II}_0 = \Phi + \overline{\Phi}$, this satisfies the Codazzi equations due to 6.1. The Codazzi equations for $\mathbf{II}$ are equivalent to the same equations for $\mathbf{II}_0$. In fact, if $\nabla$ is the Levi-Civita connection of the metric $\mathbf{I}$, we recall that $(\nabla_X \mathbf{I})(Y, Z) = 0$

$$\begin{aligned} (\nabla_X \mathbf{II})(Y, Z) &= (\nabla_X (H\mathbf{I} + \mathbf{II}_0))(Y, Z) \\ [H \text{ is constant, } \nabla \mathbf{I} = 0] \qquad &= (\nabla_X \mathbf{II}_0)(Y, Z) \ . \end{aligned}$$

Hence the Codazzi equation can be rewritten exclusively in terms of the traceless part

$$(\nabla_X \mathbf{II})(Y, Z) = (\nabla_Z \mathbf{II})(X, Y) \quad \Leftrightarrow \quad (\nabla_X \mathbf{II}_0)(Y, Z) = (\nabla_Z \mathbf{II}_0)(X, Y) \ .$$

Since we know that $\mathbf{II}_0$ satisfies the Codazzi equations, also $\mathbf{II}$ does. Regarding Gauss' equation, by Theorem 4.25, it is equivalent to $w$ solving the equation

$$\Delta_p w = e^{2w} - \|\Phi\|^2 e^{-2w} - 1 \ ,$$

which we assumed to be true by claim.

Hence $\mathbf{I}$ and$\mathbf{II}$ satisfy the Gauss-Codazzi equations and therefore we can apply Bonnet's Theorem 3.1, finding for every point of $\mathbb{D}$ a local immersion in $\mathbb{M}^{2,1}$, whose first and second fundamental forms are $\mathbf{I}$ and $\mathbf{II}$. Since $\mathbb{D}$ is simply connected, we can glue these local immersions to a global one

$$\iota : \mathbb{D} \to \mathbb{M}^{2,1},$$

unique up to post-composition with an isometry of $\mathbb{M}^{2,1}$.
The induced metric of $\iota$ is $\mathbf{I} = e^{2w} ds_p^2$, hence complete. Moreover the mean curvature is identically 1, for how we constructed the forms $\mathbf{I}$ and $\mathbf{II}$. Therefore the image surface is a complete spacelike surface in $\mathbb{M}^{2,1}$ with constant mean curvature 1. Then, by Proposition 5.2, it is an entire graph, hence a hyperbolic CMC-1 cut.

Finally, by construction, the Hopf differential of the Gauss map of the surface we just found is $\Phi$. Therefore

$$\mathscr{A}([S]) = [\Phi],$$

and $\mathscr{A}$ is surjective.

**3.** Injectivity:
Let $[S_1], [S_2] \in \mathbf{E}$ and assume that

$$\mathscr{A}([S_1]) = \mathscr{A}([S_2]).$$

We want to show that $[S_1] = [S_2]$.
Let us choose some conformal parametrizations

$$\sigma_i : \mathbb{D} \to S_i, \qquad i = 1, 2,$$

and define

$$\Phi_i = \varphi_i(z)\, dz^2$$

the corresponding quadratic differentials of their Gauss maps. Since we assumed that $\mathscr{A}([S_1]) = \mathscr{A}([S_2])$, up to a change of parameter we may assume that

$$\Phi_1 = \Phi_2 =: \Phi.$$

Then we define $w_i$ be the function associated to $S_i$ as in Theorem 5.5. Again from the theorem, we have that

$$\mathbf{I}_i = e^{2w_i} ds_p^2$$

is the first fundamental form of $S_i$ read through $\sigma_i$. Since both surfaces have Hopf differential $\Phi$, the functions $w_1$ and $w_2$ solve the same equation

$$\Delta_p w = e^{2w} - \|\Phi\|^2 e^{-2w} - 1,$$

and the metrics

$$e^{2w_1} ds_p^2, \qquad e^{2w_2} ds_p^2$$

are complete. By the Claim 6.3, it follows that

$$w_1 = w_2.$$

Hence $S_1$ and $S_2$ have the same first fundamental form. Moreover, the second fundamental forms are determined by $w$ and $\Phi$ through the relations

$$\begin{cases} \frac{1}{2}(h_{11} + h_{22}) & = e^{2\tilde{w}}, \\ h_{11} - h_{22} & = 2\Re(\varphi), \\ h_{12} & = -\Im(\varphi), \end{cases}$$

so that they also coincide.
Therefore the two surfaces have the same first and second fundamental forms on $\mathbb{D}$ and by applying Bonnet's Theorem 3.1, the corresponding immersions differ by a post-composition with an isometry of $\mathbb{M}^{2,1}$. In particular, $S_1$ and $S_2$ are isometric, hence

$$[S_1] = [S_2]$$

in $\mathbf{E}$. Which proves that $\mathscr{A}$ is injective.

$\square$

# Chapter 7

# Uniqueness for the PDE

We now begin the proof of Claim 6.3. Recall that, if $\Phi$ is a holomorphic quadratic differential on $\mathbb{D}$ and we set

$$q = \|\Phi\|,$$

the claim asserts the existence and uniqueness of a smooth function $w : \mathbb{D} \to \mathbb{R}$ satisfying

$$\Delta_p w = e^{2w} - q^2 e^{-2w} - 1, \tag{7.1}$$

such that the metric $e^{2w} ds_p^2$ is complete and

$$e^{2w} - q^2 e^{-2w} > 0 \tag{7.2}$$

on $\mathbb{D}$. In this section we prove uniqueness.

## 7.1 A generalized maximum principle

Let $M$ be a complete Riemannian manifold of dimension $n$, and fix a point $p \in M$. Assume moreover that $M$ is simply connected and has nonpositive sectional curvature. This assumption is stronger than what is strictly needed in this section, but greatly simplifies the discussion and is satisfied in the case of a complete space-like surface of $\mathbb{M}^{2,1}$. By the Cartan–Hadamard theorem, the exponential map

$$\exp_p : T_p M \to M$$

is a diffeomorphism. In particular, for every $x \in M$ there exists a unique geodesic joining $p$ to $x$. Define

$$r(x) := d(x, p).$$

We will need the following estimate on the Laplacian of $r$.

**Lemma 7.1.** *Assume that the Ricci curvature of $M$ is bounded below. Then*

*there exists a constant $C > 0$ such that*

$$\Delta r(x) \leq C$$

*for every $x \in M$ with $r(x) \geq 1$.*

*Proof.* We prove the statement in the case $\dim M = 2$, which is the only one needed in what follows.
Let us fix an orthonormal basis $(e_1, e_2)$ of $T_pM$, and consider the polar coordinates

$$\Phi(r, \theta) = \exp_p(r \cos\theta \, e_1 + r \sin\theta \, e_2).$$

We denote by $\partial_r$ and $\partial_\theta$ the corresponding coordinate vector fields. It is a well known fact (see for example [2]) that using these coordinates we can express the metric as

$$g = dr^2 + u(r, \theta)^2 \, d\theta^2$$

for a certain $u$, and that the following equation holds

$$u_{rr} = -K(r, \theta) u \ . \tag{7.3}$$

Now we observe that, by definition

$$\nabla r = (dr)^\sharp = \partial_r \ . \tag{7.4}$$

Let us fix $E_1$, $E_2$ an orthonormal frame. A suitable choice is $E_1 = \partial_r$ and $E_2 = \frac{1}{u}\partial_\theta$. Then we have

$$\begin{aligned} \Delta r &= \nabla \cdot \nabla r \\ \text{[(7.4)]} \quad &= \nabla \cdot \partial_r \\ &= \langle \nabla_{E_1} \partial_r, E_1 \rangle + \langle \nabla_{E_2} \partial_r, E_2 \rangle \\ &= \langle \nabla_{\partial_r} \partial_r, \partial_r \rangle + \frac{1}{u^2} \langle \nabla_{\partial_\theta} \partial_r, \partial_\theta \rangle \ . \end{aligned} \tag{7.5}$$

Now we observe that since at $\theta$ fixed, the map $r \mapsto \Phi(r, \theta)$ is a geodesic, we have $\nabla_{\partial_r} \partial_r = 0$. Then substituting in (7.5) we obtain

$$\Delta r = \frac{1}{u^2} \langle \nabla_{\partial_\theta} \partial_r, \partial_\theta \rangle \ . \tag{7.6}$$

We make two more observations. First we recall that

$$|\partial_\theta|^2 = \langle \partial_\theta, \partial_\theta \rangle = u^2 \ .$$

And then we observe that

$$\begin{aligned} \partial_r |\partial_\theta|^2 &= \partial_r (u^2) \\ &= 2u \partial_r (u) \\ &= 2u u_r \ . \end{aligned}$$

Continuing in (7.6), we obtain

$$\begin{aligned}
\Delta r &= \frac{1}{u^2}\left\langle \nabla_{\partial_\theta}\partial_r, \partial_\theta\right\rangle \\
[\nabla \text{ Levi-Civita}] \quad &= \frac{1}{u^2}\left\langle \nabla_{\partial_r}\partial_\theta, \partial_\theta\right\rangle \\
&= \frac{1}{2u^2}\partial_r\left\langle \partial_\theta, \partial_\theta\right\rangle \\
&= \frac{1}{2u^2}\partial_r|\partial_\theta|^2 \\
&= \frac{1}{2u^2}2uu_r \\
&= \frac{u_r}{u}\ .
\end{aligned}$$

Now set $f = \dfrac{u_r}{u} = \Delta r$. We have

$$\begin{aligned}
f_r &= \partial_r\left(\frac{u_r}{u}\right) \\
&= \frac{u_{rr}}{u} - \frac{u_r^2}{u^2} \\
&= \frac{u_{rr}}{u} - f^2 \\
[(7.3)] \quad &= -K - f^2\ .
\end{aligned}$$

Since the Gaussian curvature is bounded from below, there exists a constant $A_0 \geq 0$ such that

$$K \geq -A_0.$$

Hence

$$f_r \leq A_0 - f^2.\ . \tag{7.7}$$

Moreover, since $f$ is continuous on the compact set $\{r = 1\}$, we can choose a constant $A \geq A_0$ sufficiently large such that

$$f(1,\theta) \leq \sqrt{A} + 1 \qquad \text{for every } \theta.$$

Therefore

$$f_r \leq A - f^2 \qquad \text{and} \qquad f(1,\theta) \leq \sqrt{A} + 1$$

for every $\theta$.

We claim that

$$f(r,\theta) \leq \sqrt{A} + 1 \qquad \text{for every } r \geq 1 \text{ and every } \theta.$$

Let us fix a $\theta$. If this were false, there would exist an $r_0 > 1$ such that for every $r \in [1, r_0)$

$$f(r_0,\theta) = \sqrt{A} + 1, \qquad f(r,\theta) < \sqrt{A} + 1\ . \tag{7.8}$$

Since $r_0$ is the first value of $r$ such that $f(r,\theta) = \sqrt{A}+1$, for every $r < r_0$ sufficiently close to $r_0$ we have

$$f(r,\theta) < f(r_0,\theta).$$

Recall that

$$\partial_r f(r,\theta) = \lim_{r\to r_0^-} \frac{f(r,\theta)-f(r_0,\theta)}{r-r_0} \ .$$

But inside the limit, both the numerator and the denominator are positive for $r$ close enough to $r_0$, due to (7.8). Hence

$$f_r(r_0,\theta) \geq 0 \ .$$

On the other hand, from 7.7 we get

$$f_r(r_0,\theta) \leq A - f(r_0,\theta)^2 = A - (\sqrt{A}+1)^2 < 0,$$

which is a contradiction. Hence

$$f(r,\theta) \leq \sqrt{A}+1$$

for every $r \geq 1$ and every $\theta$.

Finally we conclude the proof by recalling that $f = \Delta r$, so that

$$\Delta r(x) \leq \sqrt{A}+1$$

for every $x$ such that $r(x) \geq 1$. □

We are ready to state and prove the maximum principle.

**Theorem 7.2.** *Let $M$ be a complete, simply connected Riemannian manifold of nonpositive sectional curvature, with Ricci curvature bounded from below. Let $f \in C^2(M)$ be bounded from above. Fix $p \in M$, and let $r(x) = d(p,x)$. Then there exist a constant $C > 0$ and a sequence $(x_k)_k \subset M$ such that*

$$f(x_k) \to \sup_M f, \qquad |\nabla f|(x_k) \leq \frac{C}{k}, \qquad \Delta f(x_k) \leq \frac{C}{k}.$$

*In particular,*

$$\limsup_{k\to\infty} \Delta f(x_k) \leq 0.$$

*Proof.* Let us define the auxiliary functions $u(x) = (f(x)-f(p)+1)$ and $L(x) = \log(r^2(x)+2)$. Then let us fix a positive integer $k$ and define the function $g$ as

$$g(x) = \frac{f(x)-f(p)+1}{(\log(r^2(x)+2)^{\frac{1}{k}}} = \frac{u(x)}{L(x)^{\frac{1}{k}}} \ .$$

Let us observe that $g(p) = \frac{1}{\log(2)^{\frac{1}{k}}} > 0$ and that $\lim_{r(x)\to\infty} g(x) = 0$. Since $g$ is

continuous, this means that it attains its maximum at a certain point $x_k$.

First, let us show that we can assume that $r(x_k) \to \infty$. If that is not the case, up to taking a subsequence, $r(x_k)$ is bounded. Then we can find a closed geodesic ball $\overline{B}(p;\varrho)$ with $\varrho > 0$ that contains $x_k$ for every $k$ sufficiently big. Since $M$ is complete, the Hopf Rinow theorem implies that $\overline{B}(p;\varrho)$ is compact. Therefore, after passing to a subsequence, we may assume that

$$x_k \to x_0 \quad .$$

But we know that $f(x_k) \to \sup f$ and that $f$ is continuous, which implies that $x_0$ is a maximum point.
This entails that

$$\nabla f(x_0) = 0 \qquad \text{and} \qquad \Delta f(x_0) \leq 0 \quad .$$

Which verifies trivially the thesis by setting $x_k \equiv x_0$. Thus we assume, from now on, that $r(x_k) \to \infty$.

Let us prove first that

$$\limsup_{k\to\infty} f(x_k) = \sup_M f \quad . \tag{7.9}$$

Assume it by contradiction to be false. Then by definition of sup and lim sup there exists a $\delta > 0$ and a $\bar{x} \in M$ such that for $k$ sufficiently big,

$$f(x_k) < f(\bar{x}) - \delta \ , \qquad \text{equivalently,} \qquad f(x_k) + \delta < f(\bar{x}) \quad . \tag{7.10}$$

Since we assumed that $r(x_k) \overset{k\to\infty}{\longrightarrow} \infty$, then we have, for $k$ sufficiently large, $r(x_k) > r(\bar{x})$ and $f(\bar{x}) > f(x_k)$ so that

$$g(\bar{x}) = \frac{f(\bar{x}) - f(p) + 1}{(\log(r^2(\bar{x}) + 2)^{\frac{1}{k}}} > \frac{f(x_k) - f(p) + 1}{(\log(r^2(x_k) + 2)^{\frac{1}{k}}} = g(x_k) \quad .$$

Which contradicts the maximality of $x_k$. It is obvious that, up to taking a subsequence, we can assume $f(x_k) \to \sup_M f$.

Let us now focus on the two estimates. As we already mentioned, the Cartan-Hadamard $M$ theorem guarantees that the function $d(p,x)^2 = r(x)^2$ is differentiable in $x_k$.
We can easily calculate the gradient of $g$,

$$\nabla g(x) = \frac{\nabla f(x)}{\log(r(x)^2 + 2)^{1/k}} - \frac{2(f(x) - f(p) + 1)r(x)}{k(r(x)^2 + 2)\log(r(x)^2 + 2)^{1+1/k}} \nabla r(x) \quad .$$

Since $x_k$ is a maximum point of $g$, we have $\nabla g(x_k) = 0$. Hence

$$\nabla f(x_k) = \frac{2(f(x_k) - f(p) + 1)\, r(x_k)}{k\,(r^2(x_k) + 2)\,\log(r^2(x_k) + 2)} \nabla r(x_k)\ , \tag{7.11}$$

$$|\nabla f|(x_k) = \frac{2(f(x_k) - f(p) + 1)\, r(x_k)}{k\,(r^2(x_k) + 2)\,\log(r^2(x_k) + 2)}\ , \tag{7.12}$$

where we used $|\nabla r(x_k)| = 1$.
We then observe, since $f$ is bounded from above, also $u$ is

$$u(x) = (f(x) - f(p) + 1) \leq \sup_M f - f(p) + 1 =: A.$$

Also, it is not difficult to show that the real function satisfies

$$\frac{2x}{(x^2 + 2)\log(x^2 + 2)} \leq 2\ , \quad . \tag{7.13}$$

so that by substituting in (7.12) we get the desired estimate

$$|\nabla f|(x_k) \leq \frac{2A}{k} \quad .$$

Let us now focus on the estimate on the Laplacian. We can find the expression of the Laplacian of $g = u/L^{\frac{1}{k}}$. First we observe that

$$\begin{aligned}
\Delta g &= \nabla \cdot \nabla \left( \frac{u}{L^{\frac{1}{k}}} \right) \\
&= \nabla \cdot \left( \nabla u L^{-\frac{1}{k}} + u \nabla (L^{-\frac{1}{k}}) \right) \\
&= \frac{\Delta u}{L^{\frac{1}{k}}} + 2 \left\langle \nabla u, \nabla (L^{-\frac{1}{k}}) \right\rangle + u \Delta (L^{-\frac{1}{k}}) \quad ,
\end{aligned} \tag{7.14}$$

then we calculate the involved quantities. First,

$$\nabla u = \nabla f, \qquad \nabla L = \frac{2r\nabla r}{r^2 + 2}, \qquad \nabla \left( \frac{1}{r^2 + 2} \right) = -\frac{2r\nabla r}{(r^2 + 2)^2},$$

$$\nabla \left( L^{-\frac{1}{k}} \right) = -\frac{1}{k} \frac{\nabla L}{L^{\frac{1}{k}+1}} = -\frac{1}{k} \frac{2r\nabla r}{(r^2 + 2) L^{\frac{1}{k}+1}},$$

$$\nabla \left( L^{-\frac{1}{k}-1} \right) = \left( -\frac{1}{k} - 1 \right) \frac{\nabla L}{L^{\frac{1}{k}+2}} = -\frac{k+1}{k} \frac{2r\nabla r}{(r^2 + 2) L^{\frac{1}{k}+2}},$$

$$\Delta L = \nabla \cdot \left( \frac{2r\nabla r}{r^2 + 2} \right) = \frac{2|\nabla r|^2}{r^2 + 2} + \frac{2r\Delta r}{r^2 + 2} - \frac{4r^2 |\nabla r|^2}{(r^2 + 2)^2} \quad ,$$

and then

$$
\begin{aligned}
\Delta\left(L^{-\frac{1}{k}}\right) &= \nabla\cdot\nabla L^{-\frac{1}{k}} \\
&= -\frac{1}{k}\nabla\cdot\left(\frac{\nabla L}{L^{\frac{1}{k}+1}}\right) \\
&= -\frac{1}{k}\frac{\Delta L}{L^{\frac{1}{k}+1}} - \frac{1}{k}\left\langle \nabla L, \nabla\left(L^{-\frac{1}{k}-1}\right)\right\rangle \\
&= -\frac{1}{k}\left(\frac{2|\nabla r|^2}{r^2+2} + \frac{2r\Delta r}{r^2+2} - \frac{4r^2|\nabla r|^2}{(r^2+2)^2}\right)\frac{1}{L^{\frac{1}{k}+1}} + \frac{k+1}{k^2}\left\langle \frac{2r\nabla r}{r^2+2}, \frac{2r\nabla r}{(r^2+2)L^{\frac{1}{k}+2}}\right\rangle \\
&= -\frac{2}{k}\frac{1}{(r^2+2)L^{\frac{1}{k}+1}} - \frac{2}{k}\frac{r\Delta r}{(r^2+2)L^{\frac{1}{k}+1}} + \frac{4}{k}\frac{r^2}{(r^2+2)^2L^{\frac{1}{k}+1}} + \frac{4(k+1)}{k^2}\frac{r^2}{(r^2+2)^2L^{\frac{1}{k}+2}} \ .
\end{aligned}
$$

Hence, by substituting in (7.14) and simplifying, we obtain

$$
\begin{aligned}
\Delta g = \frac{\Delta f}{L^{1/k}} &- \frac{4r}{k(r^2+2)L^{1+1/k}}\langle\nabla f, \nabla r\rangle - \frac{2u}{k(r^2+2)L^{1+1/k}} \\
&- \frac{2ur\,\Delta r}{k(r^2+2)L^{1+1/k}} + \frac{4ur^2}{k(r^2+2)^2L^{1+1/k}} + \frac{4(k+1)ur^2}{k^2(r^2+2)^2L^{2+1/k}} \ .
\end{aligned}
$$

Since $x_k$ is a maximum point of $g$, we have $\Delta g(x_k) \le 0$. Using the expression for $\Delta g$, the identity $\nabla g(x_k) = 0$, and

$$
\langle\nabla f, \nabla r\rangle(x_k) = \frac{2u(x_k)r(x_k)}{k(r^2(x_k)+2)\log(r^2(x_k)+2)} \ ,
$$

which we have derived from (7.11), we obtain

$$
\begin{aligned}
\Delta f(x_k) \le \frac{2u(x_k)\big(1 + r(x_k)\Delta r(x_k)\big)}{k(r^2(x_k)+2)\log(r^2(x_k)+2)} &- \frac{4u(x_k)r^2(x_k)}{k(r^2(x_k)+2)^2\log(r^2(x_k)+2)} \\
&- \frac{4(k-1)u(x_k)r^2(x_k)}{k^2(r^2(x_k)+2)^2\log^2(r^2(x_k)+2)} \ .
\end{aligned}
$$

In particular, by discarding the last two nonpositive terms, we get

$$
\Delta f(x_k) \le \frac{2u(x_k)\big(1 + r(x_k)\Delta r(x_k)\big)}{k(r^2(x_k)+2)\log(r^2(x_k)+2)} \ . \tag{7.15}
$$

First, it is easy to see that the following real function is bounded by a constant $C > 0$

$$
\frac{1}{(x^2+2)\log(x^2+2)} \le C \ . \tag{7.16}
$$

Then it follows from (7.15)

$$\begin{aligned}
\Delta f(x_k) &\leq \frac{2u(x_k)(1+r(x_k)\Delta r(x_k))}{k(r^2(x_k)+2)\log(r^2(x_k)+2)} \\
[u \leq A] \qquad &\leq \frac{2A}{k}\left(\frac{1}{(r^2(x_k)+2)\log(r^2(x_k)+2)} + \frac{r(x_k)\Delta r(x_k)}{(r^2(x_k)+2)\log(r^2(x_k)+2)}\right) \\
[(7.13),\ (7.16)] \qquad &\leq \frac{2A}{k}\left(C + 2\Delta r(x_k)\right) \quad . \qquad (7.17)
\end{aligned}$$

We now apply the estimate of Lemma 7.1. We can do this because under the assumption that $r(x_k) \to \infty$ we have $r(x_k) \geq 1$ for $k$ sufficienly large, so that there exists a constant $C > 0$ such that

$$\Delta r(x) \leq C \ .$$

Therefore we conclude in (7.17) that there is a constant $\tilde{C}$ for which

$$\Delta f(x_k) \leq \frac{\tilde{C}}{k} \ .$$

□

### 7.1.1 A boundedness result

We will also use this theorem to obtain a boundedness result for a certain differential inequality. We first make a few preliminary observations, and then state the theorem at the end of the section.

Let $M$ be a complete Riemannian manifold with Ricci curvature bounded from below, and let $u \in C^2(M)$ satisfy

$$\Delta u \geq f(u) \quad , \qquad (7.18)$$

where the assumptions on $f$ will be specified later.
Now let $F \in C^2(\mathbb{R})$ be a positive function, whose precise form will also be given later, which we require to be strictly increasing or in other terms to have a strictly positive derivative. Since $F > 0$, the function $-1/F$ is bounded from above. We may therefore apply Theorem 7.2 to $-1/F(u)$ and obtain a sequence $(x_k)_k$ such that

$$\lim_{k\to\infty} \frac{1}{F(u(x_k))} = \inf \frac{1}{F(u)} = -\sup\left(-\frac{1}{F(u)}\right). \qquad (7.19)$$

Moreover, the corresponding gradient and Laplacian estimates hold:

$$\frac{|F'(u(x_k))|\,|\nabla u|(x_k)}{F(u(x_k))^2} \leq \frac{1}{k}, \qquad (7.20)$$

$$-\frac{F''(u(x_k))\,|\nabla u|^2(x_k)}{F(u(x_k))^2} - \frac{F'(u(x_k))\,\Delta u(x_k)}{F(u(x_k))^2} + \frac{2(F'(u(x_k)))^2\,|\nabla u|^2(x_k)}{F(u(x_k))^3} \geq -\frac{1}{k}\ . \tag{7.21}$$

For simplicity, from now on we evaluate all quantities at $x_k$.
Then, multiplying (7.21) by $\dfrac{F'(u)^2}{F(u)^2|F''(u)|}$, we obtain

$$\begin{aligned} -\ \frac{F''(u)}{|F''(u)|}\frac{F'(u)^2|\nabla u|^2}{F(u)^4} \ -\ \frac{F'(u)^3}{F(u)^4}\frac{\Delta u}{|F''(u)|} \\ +\ \frac{2F'(u)^2}{F(u)|F''(u)|}\frac{F'(u)^2|\nabla u|^2}{F(u)^4} \geq -\frac{F'(u)^2}{kF(u)^2|F''(u)|}\ . \end{aligned} \tag{7.22}$$

We now estimate the terms on the left-hand side of (7.22). First,

$$-\frac{F'(u)^3}{F(u)^4}\frac{\Delta u}{|F''(u)|} \leq -\frac{F'(u)^3}{F(u)^4}\frac{f(u)}{|F''(u)|}\ . \qquad \text{[by (7.18), } F' > 0\text{]}$$

Moreover,

$$-\ \frac{F''(u)}{|F''(u)|}\frac{F'(u)^2|\nabla u|^2}{F(u)^4} \leq \left|\frac{F'(u)^2|\nabla u|^2}{F(u)^4}\right| \leq \frac{1}{k^2}\ . \qquad \text{[by (7.20)]}$$

Also,

$$\begin{aligned} \frac{2F'(u)^2}{F(u)|F''(u)|}\frac{F'(u)^2|\nabla u|^2}{F(u)^4} &\leq 2\left|\frac{F'(u)^2}{F(u)|F''(u)|}\right|\left|\frac{F'(u)^2|\nabla u|^2}{F(u)^4}\right| \\ &\leq \frac{2}{k^2}\left|\frac{F'(u)^2}{F(u)|F''(u)|}\right|\ . \qquad \text{[by (7.20)]} \end{aligned}$$

Therefore, from (7.22) we obtain

$$\frac{1}{k^2}\ -\ \frac{F'(u)^3 f(u)}{F(u)^4|F''(u)|}\ +\ \frac{2}{k^2}\left|\frac{F'(u)^2}{F(u)|F''(u)|}\right| \geq -\frac{1}{k}\frac{F'(u)^2}{F(u)^2|F''(u)|}\ . \tag{7.23}$$

Suppose now that $F$ satisfies

$$\limsup_{t\to\infty}\frac{|F'(t)|^2}{F(t)\,|F''(t)|} < \infty\ , \tag{7.24}$$

and

$$\liminf_{t\to\infty}\frac{F'(t)^3 f(t)}{F(t)^4\,|F''(t)|} > 0\ . \tag{7.25}$$

We claim that this implies

$$\limsup_{k\to\infty} u(x_k) < \infty\ .$$

Indeed, suppose by contradiction that $\limsup_{k\to\infty} u(x_k) = \infty$. Passing to a subsequence if necessary we may assume that

$$u(x_k) \to \infty \quad .$$

Taking lim sup in (7.23), and using (7.24) together with the fact that later our choice of $F$ will satisfy $F \geq 1$, we obtain

$$-\liminf_{k\to\infty} \frac{F'(u(x_k))^3 f(u(x_k))}{F(u(x_k))^4 |F''(u(x_k))|} \geq 0 \quad ,$$

that is,

$$\liminf_{k\to\infty} \frac{F'(u(x_k))^3 f(u(x_k))}{F(u(x_k))^4 |F''(u(x_k))|} \leq 0 \quad ,$$

which contradicts (7.25).

So we have proved so far that

$$\limsup_{k\to\infty} u(x_k) < \infty \quad .$$

Passing to a subsequence if necessary, we may therefore assume that

$$u(x_k) \to \ell$$

for some $\ell \in \mathbb{R}$.

Let us now show that $u$ is bounded from above.
Since $F > 0$, from (7.19) we obtain

$$\lim_{k\to\infty} \frac{1}{F(u(x_k))} = \inf \frac{1}{F(u)} = \frac{1}{\sup_M F(u)} \quad ,$$

and therefore

$$F(u(x_k)) \to \sup F(u) \quad .$$

On the other hand, by continuity of $F$ and the fact that $u(x_k) \to \ell$, we also have

$$F(u(x_k)) \to F(\ell) \quad .$$

Hence

$$F(\ell) = \sup F(u) \quad .$$

Since $F$ is strictly increasing, for every $x \in M$ we have

$$F(u(x)) \leq F(\ell) \qquad \Longrightarrow \qquad u(x) \leq \ell \quad .$$

This means that $\ell$ is an upper bound for $u$, hence $u$ is bounded from above:

$$\sup u < \infty \quad .$$

We have therefore reduced the problem to finding a function $F$ which is positive, strictly increasing, and satisfies (7.24) and (7.25). Let us search for such a function of the form

$$F(s) = \int_a^s \left( \int_a^t g(\tau)\, d\tau \right)^{-\frac{1}{2}} dt + 1 \quad , \tag{7.26}$$

where $g$ is a positive continuous function defined on $[a, \infty)$ and such that

$$\int_b^\infty \left( \int_a^t g(\tau)\, d\tau \right)^{-\frac{1}{2}} dt < \infty$$

for some $b \geq a$. If we want $F$ to be positive, the expression in (7.26) may fail to achieve that on the whole line. So we define it on $[a, \infty)$ and then extend $F$ to all of $\mathbb{R}$ in such a way that it becomes a strictly increasing $C^2$ function.

Let us now translate conditions (7.24) and (7.25) into conditions on the function $g$. To do so, we first compute the relevant quantities.

$$F'(s) = \left( \int_a^s g(\tau)\, d\tau \right)^{-\frac{1}{2}} \quad ,$$

and therefore

$$F''(s) = -\frac{1}{2} \left( \int_a^s g(\tau)\, d\tau \right)^{-\frac{3}{2}} g(s) \quad .$$

Hence

$$\frac{|F'(t)|^2}{F(t)\,|F''(t)|} = 2\, \frac{\left( \int_a^t g(\tau)\, d\tau \right)^{1/2}}{F(t)\, g(t)} \quad ,$$

so condition (7.24) is equivalent to

$$\limsup_{t \to \infty} \frac{\left( \int_a^t g(\tau)\, d\tau \right)^{1/2}}{F(t)\, g(t)} < \infty \quad .$$

Moreover,

$$\frac{F'(t)^3 f(t)}{F(t)^4 |F''(t)|} = 2\, \frac{f(t)}{F(t)^4 g(t)} \quad ,$$

so condition (7.25) is equivalent to

$$\liminf_{t\to\infty} \frac{f(t)}{F(t)^4 g(t)} > 0 \quad .$$

Finally, we observe that since $g$ is positive, if $s \le t$, then

$$\left(\int_a^s g(\tau)\,d\tau\right)^{-\frac{1}{2}} \ge \left(\int_a^t g(\tau)\,d\tau\right)^{-\frac{1}{2}} \quad .$$

Therefore, for $t \ge b$,

$$\begin{aligned} F(t) - 1 &= \int_a^t \left(\int_a^s g(\tau)\,d\tau\right)^{-\frac{1}{2}} ds \\ &\ge \int_b^t \left(\int_a^s g(\tau)\,d\tau\right)^{-\frac{1}{2}} ds \\ &\ge \int_b^t \left(\int_a^t g(\tau)\,d\tau\right)^{-\frac{1}{2}} ds \\ &= (t-b)\left(\int_a^t g(\tau)\,d\tau\right)^{-\frac{1}{2}} \quad . \end{aligned}$$

Hence

$$F(t) \ge (t-b)\left(\int_a^t g(\tau)\,d\tau\right)^{-\frac{1}{2}} \quad .$$

This allows us to further estimate the quantity appearing in (7.24):

$$\begin{aligned} \frac{\left(\int_a^t g(\tau)\,d\tau\right)^{1/2}}{F(t)\,g(t)} &\le \frac{\left(\int_a^t g(\tau)\,d\tau\right)^{1/2}}{(t-b)\left(\int_a^t g(\tau)\,d\tau\right)^{-1/2} g(t)} \\ &= \frac{\displaystyle\int_a^t g(\tau)\,d\tau}{(t-b)g(t)} \\ &\sim \frac{\displaystyle\int_a^t g(\tau)\,d\tau}{t\,g(t)} \quad . \end{aligned}$$

It is convenient to rewrite what we found in terms of the primitive of $g$.Let

$$G(t) = \int_a^t g(\tau)\,d\tau,$$

then $G \in C^1([a,\infty))$, $G(a) = 0$, and $G'(t) = g(t) > 0$ for every $t \ge a$. In these terms, the previous discussion yields the following theorem.

**Theorem 7.3.** *Let $M$ be a complete Riemannian manifold whose Ricci curva-*

*ture is bounded from below, and let $u \in C^2(M)$ satisfy*

$$\Delta u \geq f(u) \quad .$$

*A sufficient condition for $u$ to be bounded from above on $M$ is the existence of $a \in \mathbb{R}$ and of an increasing positive function $G \in C^1([a, \infty))$ such that*

$$\begin{cases} \displaystyle\int_b^\infty G(t)^{-1/2}\, dt < \infty \text{ for some } b \geq a \\ \displaystyle\limsup_{t\to\infty} \frac{G(t)}{t\, G'(t)} < \infty \\ \displaystyle\liminf_{t\to\infty} \frac{f(t)}{G'(t)} > 0 \ . \end{cases}$$

## 7.2 The uniqueness theorem

We now prove the uniqueness part of Claim 6.3. We recall that, if

$$w : \mathbb{D} \to \mathbb{R}$$

solves the equation

$$\Delta_p w = e^{2w} - q^2 e^{-2w} - 1,$$

then the function

$$\tilde{w} = w + \log\left(\frac{2}{1 - |z|^2}\right) \tag{7.27}$$

solves an equivalent equation with respect to the Euclidean background metric. Therefore it is enough to prove uniqueness in the Euclidean formulation and then translate the conclusion back to the original one.

**Theorem 7.4.** *Let $h$ be a continuous function defined on a domain $\Omega \subset \mathbb{C}$. Let $\tilde{w}$ be a solution of*

$$\Delta \tilde{w} = e^{2\tilde{w}} - h^2 e^{-2\tilde{w}} \tag{7.28}$$

*in $\Omega$, and assume that*

$$\Delta \tilde{w} \geq 0 \quad ,$$

*and that $e^{2\tilde{w}}|dz|^2$ is a complete metric on $\Omega$. Then any other solution $v$ of the equation satisfies $v \leq \tilde{w}$.*

*Proof.* Let us define $\eta = v - \tilde{w}$. We need to prove that $\eta \leq 0$.
We have

$$\begin{aligned} \Delta\eta &= \Delta v - \Delta\tilde{w} \\ &\overset{[(7.28)]}{=} e^{2v} - h^2 e^{-2v} - e^{2\tilde{w}} + h^2 e^{-2\tilde{w}} \quad . \end{aligned} \tag{7.29}$$

Let us now try to explicit $\tilde{w}$ and $\eta$ in the RHS of (7.29),

$$e^{2v} = e^{2(\eta+\tilde{w})} = e^{2\eta}e^{2\tilde{w}} \quad .$$

Let us define the quantity $|\mu|^2 = h^2 e^{-4\tilde{w}}$, we have then

$$h^2 e^{-2v} = h^2 e^{-2(\eta+\tilde{w})} = h^2 e^{-2\tilde{w}} e^{-2\eta} = |\mu|^2 e^{2\tilde{w}} e^{-2\eta},$$

and

$$h^2 e^{-2\tilde{w}} = \left(h^2 e^{-4\tilde{w}}\right) e^{2\tilde{w}} = |\mu|^2 e^{2\tilde{w}} \quad .$$

So, by substituting what we just found in (7.29), we get

$$\begin{aligned} \Delta\eta &= e^{2\tilde{w}} e^{2\eta} - |\mu|^2 e^{2\tilde{w}} e^{-2\eta} - e^{2\tilde{w}} + |\mu|^2 e^{2\tilde{w}} \\ &= e^{2\tilde{w}}(e^{2\eta} - |\mu|^2 e^{-2\eta} - 1 + |\mu|^2) \quad . \end{aligned} \tag{7.30}$$

The Laplacian with respect to the metric $g = e^{2\tilde{w}}|dz|^2$ is simply $e^{-2\tilde{w}}\Delta$. So in (7.30) we obtain

$$\Delta_g \eta = e^{-2\tilde{w}}\Delta\eta = e^{2\eta} - |\mu|^2 e^{-2\eta} - 1 + |\mu|^2 \quad . \tag{7.31}$$

Let us now observe that $|\mu|^2 \leq 1$. Indeed, by (7.28),

$$e^{2\tilde{w}} - h^2 e^{-2\tilde{w}} \geq 0 \quad ,$$

hence

$$e^{2\tilde{w}}(1 - h^2 e^{-4\tilde{w}}) \geq 0 \quad ,$$

that is,

$$1 - |\mu|^2 \geq 0 \quad ,$$

so $|\mu|^2 \leq 1$.
Therefore, from (7.31) we get

$$\begin{aligned} \Delta_g \eta &= e^{2\eta} - |\mu|^2 e^{-2\eta} - 1 + |\mu|^2 \\ &\geq e^{2\eta} - e^{-2\eta} - 1 =: f(\eta) \quad . \end{aligned}$$

We now apply Theorem 7.3 to the function $\eta$ on the complete simply connected Riemannian surface $(\Omega, g)$. For the auxiliary function appearing in the theorem, we may take

$$\gamma(t) = t^2$$

on $[1, \infty)$. A direct computation shows that $\gamma$ satisfies the required assumptions, since

$$\int_b^\infty \left(\int_1^t \tau^2 \, d\tau\right)^{-1/2} dt < \infty, \qquad \limsup_{t\to\infty} \frac{\int_1^t \tau^2 \, d\tau}{t^3} < \infty,$$

and

$$\liminf_{t\to\infty} \frac{f(t)}{t^2} = \liminf_{t\to\infty} \frac{e^{2t} - e^{-2t} - 1}{t^2} > 0 \quad .$$

It remains only to check that the Ricci curvature of $g$ is bounded from below which, since we are on a surface, is the same as the Gaussian curvature. For the metric

$$g = e^{2\tilde{w}}|dz|^2 \quad ,$$

the Gaussian curvature is

$$\begin{aligned} K(g) &= -e^{-2\tilde{w}}\Delta\tilde{w} \\ &= -e^{-2\tilde{w}}(e^{2\tilde{w}} - h^2 e^{-2\tilde{w}}) \\ &= -1 + h^2 e^{-4\tilde{w}} \\ &= -1 + |\mu|^2 \geq -1 \quad . \end{aligned} \tag{7.32}$$

Since $|\mu|^2 \leq 1$, it follows that

$$-1 \leq K(g) \leq 0 \quad ,$$

and in particular, looking at (7.32), the curvature is bounded from below. Therefore we can apply Theorem 7.3 and conclude that $\eta$ is bounded from above.
Now we can also apply Theorem 7.2 to $\eta$, obtaining a sequence $(z_k)_k \subset \Omega$ such that

$$\eta(z_k) \to \bar{\eta} = \sup_{\Omega} \eta, \qquad \limsup_{k\to\infty} \Delta_g \eta(z_k) \leq 0 \quad .$$

Since $|\mu|^2$ is continuous and bounded, by passing to a subsequence if necessary we may also assume that

$$|\mu|^2(z_k) \to a$$

for some $a \leq 1$. Taking limits in (7.31), we obtain

$$\begin{aligned} 0 &\geq \limsup_{k\to\infty} \Delta_g \eta(z_k) \\ &= \lim_{k\to\infty} \left( e^{2\eta(z_k)} - |\mu|^2(z_k) e^{-2\eta(z_k)} - 1 + |\mu|^2(z_k) \right) \\ &= e^{2\bar{\eta}} - a e^{-2\bar{\eta}} - 1 + a \\ &=: \varphi(\bar{\eta}) \quad . \end{aligned} \tag{7.33}$$

We have defined in the last expression the function $\varphi(s) = e^{2s} - ae^{-2s} - 1 + a$, which has strictly positive derivative $\varphi'(s) = 2e^{2s} + 2ae^{-2s} > 0$, so that it is

strictly increasing. Moreover $\varphi(0) = 0$, hence (7.33) implies

$$\bar{\eta} \leq 0, \qquad \text{therefore, on } \Omega, \qquad \eta \leq 0 \quad .$$

This finally implies $v \leq \tilde{w}$. □

**Corollary 7.5.** *Let $h$ be a continuous function defined on a domain $\Omega \subset \mathbb{C}$ and let $\tilde{w}_1, \tilde{w}_2$ two solutions of*

$$\Delta \tilde{w} = e^{2\tilde{w}} - h^2 e^{-2\tilde{w}} \geq 0 \tag{7.34}$$

*in $\Omega$ such that $e^{2\tilde{w}_i}$ are complete metrics on $\Omega$. Then $\tilde{w}_1 \equiv \tilde{w}_2$.*

*Proof.* Just observe that for the previous theorem, $\tilde{w}_1 \leq \tilde{w}_2$ and $\tilde{w}_2 \leq \tilde{w}_1$. □

**Corollary 7.6.** *There is at most one solution of the equation* (7.1) *such that the associated metric is complete.*

*Proof.* Let $w_1, w_2$ be two solutions of

$$\Delta_p w = e^{2w} - q^2 e^{-2w} - 1$$

satisfying (7.2), and assume that the metrics $e^{2w_1} ds_p^2$ and $e^{2w_2} ds_p^2$ are complete.

For $i = 1, 2$, define

$$\tilde{w}_i = w_i + \log\left(\frac{2}{1 - |z|^2}\right) \quad .$$

Then

$$e^{2\tilde{w}_i} |dz|^2 = e^{2w_i} ds_p^2,$$

hence the metric $e^{2\tilde{w}_i}|dz|^2$ is complete, as it is the same metric as $e^{2w_i} ds_p^2$, assumed to be complete by hypothesis.

Now, set

$$h(z) = \frac{4q(z)}{(1 - |z|^2)^2}.$$

By recalling the discussion in Section 4.6 the equation can be rewritten with the Euclidean background metric so that $\tilde{w}_i$ solves

$$\Delta \tilde{w}_i = e^{2\tilde{w}_i} - h^2 e^{-2\tilde{w}_i}.$$

Moreover, condition (7.2) is equivalent to

$$e^{2\tilde{w}_i} - h^2 e^{-2\tilde{w}_i} > 0,$$

and so

$$\Delta \tilde{w}_i \geq 0.$$

Therefore $\tilde{w}_1$ and $\tilde{w}_2$ satisfy the assumptions of Corollary 7.5, so that

$$\tilde{w}_1 = \tilde{w}_2.$$

Which finally implies that

$$w_1 = w_2 \quad .$$

$\square$

# Chapter 8

# Existence for the PDE

## 8.1 Analytic preliminaries

In this chapter we will recall some essential concepts of functional analysis, that will be used extensively in the proof of the existence theorem. We report these results in this section as they come useful only now.

Let us fix $\Omega \subset \mathbb{R}^n$ an open set. We assume the reader to be familiar with the Sobolev spaces $W^{k,p}$ and the spaces $C^{k,\alpha}$, but we will recall some basic information.

We will use the Sobolev immersion theorem, in the following form, so it will be useful to recall its statement.

**Theorem 8.1** (Sobolev embedding on domains satisfying the cone condition)**.** *Let $\Omega$ be a bounded domain with regular boundary. Let $\ell \in \mathbb{N}$ and $1 \leq p < \infty$. If*

$$m < \ell - \frac{n}{p},$$

*then there is a continuous embedding*

$$W^{\ell,p}(\Omega) \hookrightarrow C^m(\overline{\Omega}).$$

We shall use only the following special case of the Ascoli-Arzelà theorem.

**Theorem 8.2** (Ascoli-Arzelà)**.** *Suppose $\Omega$ to be bounded, and let $(f_j)_j$ be a sequence bounded in $C^{m+1}(\overline{\Omega})$. Then there exists a subsequence $(f_{j_k})_k$ and a function $f \in C^m(\overline{\Omega})$ such that*

$$f_{j_k} \to f \qquad \text{in } C^m(\overline{\Omega}).$$

Then we will need the following notion of convergence, which will be used show that the limit of a sequence of solutions of an equation is itself a solution.

**Definition 8.3.** A sequence $(u_j)_j \subset C^2(\Omega)$ is said to converge to $u$ in $C^2_{\text{loc}}(\Omega)$

if for every compact set $K \subset \Omega$,

$$\|u_j - u\|_{C^2(K)} \to 0 \ .$$

Then, we will need the following theorem, whose proof can be found in [6]

**Theorem 8.4** (Elliptic interior $L^2$-estimate)**.** *Let $\Omega', \Omega \subset \mathbb{R}^n$ be bounded domains with $\overline{\Omega'} \subset \Omega$, and let $\ell \in \mathbb{N} \cup \{0\}$. Suppose that $u \in H^{\ell+2}_{\mathrm{loc}}(\Omega)$ satisfies*

$$\Delta u = f$$

*on $\Omega$, with $f \in H^{\ell}(\Omega)$. Then $u \in H^{\ell+2}(\Omega')$ and there exists a constant $C = C(\ell, \Omega', \Omega)$ such that*

$$\|u\|_{H^{\ell+2}(\Omega')} \leq C\Big(\|u\|_{H^1(\Omega)} + \|f\|_{H^{\ell}(\Omega)}\Big).$$

Finally, we might recall the statement of the method for sub- and super-solutions, in the form that can be found in [10].

**Theorem 8.5.** *Let $M$ be a complete Riemannian manifold, and let $F(x, u)$ be a $C^\infty$ function on $M \times \mathbb{R}$ such that*

$$\frac{\partial F}{\partial u} > 0.$$

*If there exist two functions $\underline{\psi} \leq \overline{\psi} \in C^0(M) \cap H^1(M)$ such that*

$$\begin{cases} \Delta\overline{\psi} \leq F(x, \overline{\psi}) \\ \Delta\underline{\psi} \geq F(x, \underline{\psi}) \end{cases} \quad \textit{weakly} \ ,$$

*then there is a certain $u \in C^\infty(M)$ such that*

$$\Delta u = F(x, u)$$

*and*

$$\underline{\psi} \leq u \leq \overline{\psi}.$$

*We call $\underline{\psi}$ and $\overline{\psi}$ respectively the sub- and super-solutions.*

## 8.2 The bounded case

We may prove the existence theorem with the stronger hypothesis of boundedness for the norm of the quadratic differential.

**Proposition 8.6.** *Let $\Phi$ a holomorphic quadratic differential on $\mathbb{D}$, such that $\|\Phi\|$ is bounded. Then the equation* (7.1) *has a $C^\infty$ solution $w$ that satisfies*

$$1 \leq e^{2w} \leq \frac{1 + \sqrt{1 + 4a^2}}{2} \ ,$$

*with $a = \sup \|\Phi\|$.*

*Proof.* Set

$$F(x,u) = e^{2u} - \|\Phi\|^2(x)e^{-2u} - 1.$$

Then

$$\frac{\partial F}{\partial u}(x,u) = 2e^{2u} + 2\|\Phi\|^2(x)e^{-2u} > 0,$$

so we may apply Theorem 8.5 once we find a subsolution and a supersolution.

First we claim that

$$\underline{\psi} := \frac{1}{2}\log^+ \|\Phi\|$$

is a weak subsolution, where

$$\log^+ \|\Phi\| = \max\{\log \|\Phi\|, 0\}.$$

Indeed,

$$\underline{\psi} = \sup\left\{0, \frac{1}{2}\log \|\Phi\|\right\},$$

so it is enough to show that both 0 and $\frac{1}{2}\log \|\Phi\|$ are weak subsolutions.

The constant zero is a subsolution, because

$$\Delta_p 0 - e^{2\cdot 0} + \|\Phi\|^2 e^{-2\cdot 0} + 1 = \|\Phi\|^2 \geq 0.$$

For $\frac{1}{2}\log \|\Phi\|$, recalling that $\Phi = \varphi(z)\,dz^2$ is holomorphic, we have that $\log|\varphi|$ is weakly subharmonic. Since

$$\|\Phi\| = \varrho^{-2}|\varphi|$$

with respect to the Poincaré metric, it follows that

$$\Delta_p \log \|\Phi\| = \Delta_p \log|\varphi| + 2$$

weakly. Hence

$$\begin{aligned}\Delta_p\left(\frac{1}{2}\log \|\Phi\|\right) - e^{\log \|\Phi\|} + \|\Phi\|^2 e^{-\log \|\Phi\|} + 1 &= \frac{1}{2}\Delta_p \log \|\Phi\| - \|\Phi\| + \|\Phi\| + 1 \\ &= \frac{1}{2}\Delta_p \log \|\Phi\| + 1 \\ &= \frac{1}{2}\Delta_p \log|\varphi| \geq 0\end{aligned}$$

weakly. Therefore $\underline{\psi}$ is a weak subsolution.

Now let us find a constant supersolution. Set $\overline{\psi} \equiv C$. Then we need

$$0 \geq \Delta_p C - e^{2C} + \|\Phi\|^2 e^{-2C} + 1 = -e^{2C} + \|\Phi\|^2 e^{-2C} + 1.$$

If we set $X = e^{2C}$ and $a = \sup \|\Phi\|$, it is enough to require

$$-X + a^2 X^{-1} + 1 \leq 0,$$

that is,

$$X^2 - X - a^2 \geq 0.$$

Therefore it is enough to choose

$$X = \frac{1 + \sqrt{1 + 4a^2}}{2},$$

namely

$$\overline{\psi} = \frac{1}{2} \log \left( \frac{1 + \sqrt{1 + 4a^2}}{2} \right).$$

By Theorem 8.5, there exists a smooth solution $w$ with

$$\frac{1}{2} \log^+ \|\Phi\| \leq w \leq \frac{1}{2} \log \left( \frac{1 + \sqrt{1 + 4a^2}}{2} \right).$$

Exponentiating, we obtain

$$1 \leq e^{2w} \leq \frac{1 + \sqrt{1 + 4a^2}}{2}.$$

□

## 8.3 The general case

**Theorem 8.7.** *Let $\Phi$ be a holomorphic quadratic differential on $\mathbb{D}$, then the equation*

$$\Delta \tilde{w} = e^{2\tilde{w}} - h^2 e^{-2\tilde{w}} \ .$$

*has a $C^\infty$ solution.*

*Proof.* The idea of this proof is to construct a sequence of solutions on consecutive domains that exhaust $\mathbb{D}$ and prove their convergence in a topology sufficient to obtain the equation valid for the limit.

For every positive integer $k$, denoting with $d_p(\cdot, \cdot)$ the Poincaré distance on $\mathbb{D}$, define the sequence of open bounded sets

$$\begin{aligned} D_k &:= \{z \in \mathbb{D} \mid d_p(z, 0) < k\} \\ &= \{z \in \mathbb{D} \mid |z| < R_k\} \ , \end{aligned}$$

where $R_k = \frac{e^k - 1}{e^k + 1} < 1$. Then

$$\overline{D_k} \subset D_{k+1} \text{ for every } k, \qquad \bigcup_{k=1}^{\infty} D_k = \mathbb{D} \ .$$

Every $D_k$ can be seen as a Riemannian manifold, with the metric

$$ds^2_{p,k} = \frac{4R_k^2}{(R_k^2 - |z|^2)^2}|dz|^2 = \lambda^2_{p,k}|dz|^2 \ .$$

Furthermore $\overline{D_k}$ is compact and $|\Phi|$ is smooth, then $|\Phi|$ is bounded on $\overline{D_k}$. Hence we may apply Proposition 8.6 and find, for every $k$, a function $\tilde{w}_k$ that solves

$$\begin{cases} \Delta \tilde{w}_k = e^{2\tilde{w}_k} - |\Phi|^2 e^{-2\tilde{w}_k} \\ (D_k, e^{2\tilde{w}_k}|dz|^2) \text{ is complete.} \end{cases}$$

It may be useful to define $w_k$ as the conformal factor with respect to the metric $ds^2_{p,k}$. Then

$$\begin{aligned} \tilde{w}_k &= w_k + \log(\lambda_{p,k}) \\ &= w_k + \log\left(\frac{2R_k}{R_k^2 - |z|^2}\right) \ . \end{aligned}$$

In the proof of Proposition 8.6 we have seen that $e^{2w_k} \geq 1$, this translates on $\tilde{w}_k$ to

$$e^{2\tilde{w}_k} = e^{2w_k}\frac{4R_k^2}{(R_k^2 - |z|^2)^2} \geq \frac{4R_k^2}{(R_k^2 - |z|^2)^2} \geq \frac{4}{(1 - |z|^2)^2} \ , \tag{8.1}$$

where in the last inequality we used that $\frac{R_k}{R_k^2 - r^2} \geq \frac{1}{1-r^2}$, which follows from $0 < r < R_k < 1$.

Now, since $(D_k, e^{2\tilde{w}_k}|dz|^2)$ is complete and $\tilde{w}_{k+1}|_{D_k}$ is still a solution to

$$\Delta w = e^{2\tilde{w}} - |\Phi|^2 e^{-2\tilde{w}} \text{ on } D_k \ ,$$

we can apply Theorem 7.4, and obtain that

$$\tilde{w}_{k+1} \leq \tilde{w}_k \text{ on } D_k \ . \tag{8.2}$$

This implies that $(\tilde{w}_{k+j})_j$ is a sequence of pointwise decreasing functions on $D_k$. This fact in turn implies that $(\tilde{w}_k)_k$ is a pointwise convergent sequence of functions on $\mathbb{D}$. In fact, choose $x \in \mathbb{D}$, then there exists $k$ sufficiently large such that $x \in D_k$. From what we just showed, $(\tilde{w}_{k+j}(x))_j$ is convergent, hence the sequence $(\tilde{w}_k(x))_k$, which is well defined for $k$ sufficiently large, is convergent as well. Let $\tilde{w}$ be its pointwise limit.

The pointwise convergence is obviously not sufficient to show that the limit satisfies the equation. For that, we will prove the convergence in $C^2_{\text{loc}}(\mathbb{D})$ for a subsequence. Since the pointwise limit has already been found, this will be sufficient to conclude that $\tilde{w}$ satisfies the equation.

Since we already have a sequence of domains that invades $\mathbb{D}$, we may prove the $C^2$ convergence on a certain $D_k$. Then, to get the full $C^2_{\text{loc}}$ convergence on $\mathbb{D}$ we will apply a diagonal argument at the end. Instead, in order to have the $C^2_{\text{loc}}$ convergence, we may prove the boundedness in the more regular space $C^3$ and apply the Ascoli-Arzelà theorem.

So, let us fix a positive integer $k$ and consider the sequence of functions $(\tilde{w}_{k+j+2})_{j=1}^{\infty}$ on $D_k$. Since $D_k$ is a disk, we can choose a smooth bounded domain $B_k$ such that

$$\overline{D_k} \subset B_k \subset \overline{B_k} \subset D_{k+1} \ .$$

We will now procede to apply a series of inequalities to $\|\tilde{w}_{k+j+2}\|_{C^3(\overline{D_k})}$ in order to obtain finally an estimate independent of $j$.

Since $B_k$ has sufficiently regular boundary, we can apply the Sobolev immersion $H^\ell \hookrightarrow C^m$, with indexes $\ell = 5$, $m = 3$, together with the obvious immersion $C^3(B_k) \hookrightarrow C^3(\overline{D_k})$, and obtain

$$\|u\|_{C^3(\overline{D_k})} \leq \|u\|_{C^3(B_k)} \leq C(k)\|u\|_{H^5(B_k)} \ . \tag{8.3}$$

We also recall the elliptic interior $L^2$-estimate: given bounded domains $\Omega', \Omega$ such that $\overline{\Omega'} \subset \Omega$, there is a constant $C = C(\ell, \Omega', \Omega)$ such that

$$\|u\|_{H^{\ell+2}(\Omega')} \leq C \left( \|u\|_{H^1(\Omega)} + \|\Delta u\|_{H^\ell(\Omega)} \right) \ .$$

Using the equation, we will show now that we can estimate $\|\tilde{w}_{k+j+2}\|_{C^3(\overline{D_k})}$ using $\|\tilde{w}_{k+j+2}\|_{H^1(D_{k+1})}$. To simplify the notation, let us denote with $v = \tilde{w}_{k+j+2}$ for the remainder of the proof.

We know that $v$ is a solution of the equation

$$\Delta v = e^{2v} - |\Phi|^2 e^{-2v} \ ,$$

from which we obtain, by taking the gradient,

$$\nabla(\Delta v) = 2(e^{2v} + |\Phi|^2 e^{-2v})\nabla v - e^{-2v}\nabla|\Phi|^2 \ .$$

From this we have, on $D_{k+1}$, a uniform bound in $j$ for $v = \tilde{w}_{k+j+2}$; in fact

- ▷ $|\Phi|$ and $\nabla|\Phi|^2$ are smooth, hence bounded on $\overline{D_{k+1}}$,
- ▷ for every $j \geq 1$, from (8.2) we have

$$v = \tilde{w}_{k+j+2} \leq \tilde{w}_{k+3} \qquad \text{on } D_{k+1},$$

- ▷ from (8.1) we have

$$e^{2v} = e^{2\tilde{w}_{k+j+2}} \geq \frac{4}{(1-|z|^2)^2} \qquad \text{on } D_{k+1}. \tag{8.4}$$

Then there exists a constant $C = C(k)$ such that, pointwise on $D_{k+1}$,

$$|\nabla(\Delta v)| \leq C(k)(|\nabla v| + 1) \ .$$

This in turn means that

$$\|\nabla(\Delta v)\|_{L^2(D_{k+1})} \leq C(k)(\|\nabla v\|_{L^2(D_{k+1})} + 1) \ . \tag{8.5}$$

Now, again from the equation, and by applying the bounds above, we obtain

$$|\Delta v| = |e^{2v} - |\Phi|^2 e^{-2v}| \leq C(k) \qquad \text{on } D_{k+1}, \tag{8.6}$$

hence

$$\|\Delta v\|_{L^2(D_{k+1})} \leq C(k) \ . \tag{8.7}$$

Therefore

$$\begin{aligned} \|\Delta v\|_{H^1(D_{k+1})} &= \|\Delta v\|_{L^2(D_{k+1})} + \|\nabla(\Delta v)\|_{L^2(D_{k+1})} \\ &\text{[(8.7),(8.5)]} \leq C(k)(1 + \|\nabla v\|_{L^2(D_{k+1})}) \\ &\leq C(k)\left(1 + \|v\|_{H^1(D_{k+1})}\right) \ . \end{aligned} \tag{8.8}$$

Choose now a smooth bounded domain $\Omega'$ such that

$$\overline{B_k} \subset \Omega' \subset \overline{\Omega'} \subset D_{k+1} \ .$$

By the interior $L^2$-estimate with $\ell = 1$, applied to the pair $(\Omega', D_{k+1})$, we obtain

$$\begin{aligned} \|v\|_{H^3(\Omega')} &\leq C(k, \Omega')\left(\|v\|_{H^1(D_{k+1})} + \|\Delta v\|_{H^1(D_{k+1})}\right) \\ &\text{[(8.8)]} \leq C(k, \Omega')\left(1 + \|v\|_{H^1(D_{k+1})}\right) \ . \end{aligned} \tag{8.9}$$

By Sobolev immersion on $\Omega'$, from (8.9) we also get a uniform $C^1$ bound for $v$ on $\Omega'$. Since $|\Phi|^2$ is smooth, by differentiating the equation up to order 3 and using the previous $C^1$ bound, we obtain

$$\|\Delta v\|_{H^3(\Omega')} \leq C(k, \Omega')\left(1 + \|v\|_{H^3(\Omega')}\right) \ . \tag{8.10}$$

Now we apply the interior $L^2$-estimate again, this time with $\ell = 3$, to the pair $(B_k, \Omega')$:

$$\begin{aligned} \|v\|_{H^5(B_k)} &\leq C(k, \Omega')\left(\|v\|_{H^1(\Omega')} + \|\Delta v\|_{H^3(\Omega')}\right) \\ &\text{[(8.10)]} \leq C(k, \Omega')\left(1 + \|v\|_{H^1(\Omega')} + \|v\|_{H^3(\Omega')}\right) \\ &\text{[(8.9)]} \leq C(k, \Omega')\left(1 + \|v\|_{H^1(D_{k+1})}\right) \ . \end{aligned} \tag{8.11}$$

So that finally it is sufficient to estimate $\|v\|_{H^1(D_{k+1})}$.

Similarly to before, $v = \tilde{w}_{k+j+2}$ solves the equation on $D_{k+2}$

$$\Delta v = e^{2v} - |\Phi|^2 e^{-2v} \quad \text{on } D_{k+2} \ .$$

Choose a positive test function $\varphi \in C_c^\infty(D_{k+2})$ such that $\varphi \equiv 1$ on $D_{k+1}$. Then

$$\int_{D_{k+2}} \varphi^2 v \Delta v = \int_{D_{k+2}} v\varphi^2 (e^{2v} - |\Phi|^2 e^{-2v}) \geq 0 \ .$$

This implies that

$$\int_{D_{k+2}} \nabla(\varphi^2 v) \cdot \nabla v = -\int_{D_{k+2}} \varphi^2 v \Delta v \leq 0 \ .$$

Hence

$$\begin{aligned} 0 &\geq \int_{D_{k+2}} \nabla(\varphi^2 v) \cdot \nabla v \\ &= \int_{D_{k+2}} 2\varphi\, v\, \nabla\varphi \cdot \nabla v + \varphi^2 |\nabla v|^2 \ , \end{aligned}$$

so that

$$\begin{aligned} \int_{D_{k+2}} \varphi^2 |\nabla v|^2 &\leq -2 \int_{D_{k+2}} \varphi v \nabla\varphi \cdot \nabla v \\ &\leq 2 \int_{D_{k+2}} \varphi v |\nabla\varphi||\nabla v| \\ &= 2 \left\langle \varphi \nabla v, v \nabla \varphi \right\rangle_{L^2(D_{k+2})} \\ &\leq 2 \|\varphi \nabla v\|_{L^2(D_{k+2})} \|v \nabla \varphi\|_{L^2(D_{k+2})} \\ &= 2 \left( \int_{D_{k+2}} \varphi^2 |\nabla v|^2 \right)^{\frac{1}{2}} \left( \int_{D_{k+2}} v^2 |\nabla \varphi|^2 \right)^{\frac{1}{2}} \ . \end{aligned}$$

This is an inequality of the form $A^2 \leq 2AB$, where

$$A = \left( \int_{D_{k+2}} \varphi^2 |\nabla v|^2 \right)^{\frac{1}{2}}, \qquad B = \left( \int_{D_{k+2}} v^2 |\nabla \varphi|^2 \right)^{\frac{1}{2}} .$$

If $A = 0$ there is nothing to prove; otherwise, dividing by $A$, we obtain $A \leq 2B$, so that

$$\int_{D_{k+2}} \varphi^2 |\nabla v|^2 = A^2 \leq 4B^2 = 4 \int_{D_{k+2}} v^2 |\nabla \varphi|^2 \ . \tag{8.12}$$

Then, remembering that $\varphi$ is a test function that depends only on $k$ and is

equal to 1 on $D_{k+1}$,

$$\begin{aligned}
\|\nabla v\|^2_{L^2(D_{k+1})} &= \int_{D_{k+1}} |\nabla v|^2 \\
&\leq \int_{D_{k+2}} \varphi^2 |\nabla v|^2 \\
[(8.12)] \quad &\leq 4 \int_{D_{k+2}} v^2 |\nabla \varphi|^2 \\
&\leq 4 \|\nabla \varphi\|^2_{L^\infty(D_{k+2})} \|v\|^2_{L^2(D_{k+2})} \\
&= C(k) \|v\|^2_{L^2(D_{k+2})} \ .
\end{aligned}$$

Then we have

$$\begin{aligned}
\|v\|_{H^1(D_{k+1})} &= \|v\|_{L^2(D_{k+1})} + \|\nabla v\|_{L^2(D_{k+1})} \\
&\leq C(k) \|v\|_{L^2(D_{k+2})} \ .
\end{aligned} \tag{8.13}$$

Since, by (8.2), we have $v = \tilde{w}_{k+j+2} \leq \tilde{w}_{k+3}$ on $D_{k+2}$, while by (8.1) we have

$$v = \tilde{w}_{k+j+2} \geq \log\left(\frac{2}{1-|z|^2}\right) \qquad \text{on } D_{k+2},$$

and both bounds depend only on $k$, there exists $C(k)$ such that

$$|v(z)| \leq C(k) \qquad \forall z \in D_{k+2}. \tag{8.14}$$

Hence

$$\|v\|_{L^2(D_{k+2})} \leq C(k). \tag{8.15}$$

Therefore

$$\begin{aligned}
\|\tilde{w}_{k+j+2}\|_{C^3(\overline{D_k})} &\overset{(8.3)}{\leq} C(k) \|\tilde{w}_{k+j+2}\|_{H^5(B_k)} \\
&\overset{(8.11)}{\leq} C(k) \big(1 + \|\tilde{w}_{k+j+2}\|_{H^1(D_{k+1})}\big) \\
&\overset{(8.13)}{\leq} C(k) \big(1 + \|\tilde{w}_{k+j+2}\|_{L^2(D_{k+2})}\big) \\
&\overset{(8.15)}{\leq} C(k).
\end{aligned}$$

So the sequence $(\tilde{w}_{k+j+2})_j$ is uniformly bounded in $C^3(\overline{D_k})$.

We may briefly illustrate the diagonal argument yielding the convergence in $C^2_{\text{loc}}$.
For every $a \in \mathbb{N}$, we proved above that the sequence $(\tilde{w}_{a+j+2})_{j=1}^\infty$ is uniformly bounded in $C^3(\overline{D_a})$.
Define $w_{0,b} := \tilde{w}_b$ and fix a positive integer index $a$. Up to taking a subsequence, we may assume that $(w_{a-1,b})_b$ is a subsequence of $(\tilde{w}_{a+j+2})_j$, hence it is bounded in $C^3(\overline{D_a})$. By the Ascoli-Arzelà theorem we can therefore extract

a subsequence $(w_{a,b})_b$ of $(w_{a-1,b})_b$ which converges in $C^2(\overline{D_a})$.
We claim that the diagonal sequence $(w_{a,a})_a$ converges in $C^2_{\text{loc}}$.
To show that, first fix an index $m$. By construction $(w_{m,b})_b$ converges in $C^2(\overline{D_m})$. Then, if $a_0 \geq m$,

$$(w_{a,a})_{a\geq a_0} \text{ is a subsequence of } (w_{m,b})_b,$$

hence $(w_{a,a})_{a\geq a_0}$ converges in $C^2(\overline{D_m})$. Since $m$ is arbitrary, and every compact set $K \subset \mathbb{D}$ is contained in $D_m$ for some $m$, the diagonal sequence converges in $C^2(K)$ for every compact set $K \subset \mathbb{D}$. Denoting by $(\tilde{w}_{k'})$ the diagonal subsequence, we have then

$$\tilde{w}_{k'} \to \tilde{w} \qquad \text{in } C^2_{\text{loc}}(\mathbb{D}),$$

and since the original sequence already converges pointwise to $\tilde{w}$, the limit of the diagonal subsequence is necessarily the same function $\tilde{w}$.

Now let $K \subset \mathbb{D}$ be a compact set. Since $\tilde{w}_{k'} \to \tilde{w}$ in $C^2(K)$, we have

$$\tilde{w}_{k'} \to \tilde{w}, \qquad \Delta\tilde{w}_{k'} \to \Delta\tilde{w}$$

uniformly on $K$. Moreover, since $|\Phi|^2$ is fixed and smooth, from the uniform convergence of $\tilde{w}_{k'}$ to $\tilde{w}$ on $K$ we also get

$$e^{2\tilde{w}_{k'}} \to e^{2\tilde{w}}, \qquad |\Phi|^2 e^{-2\tilde{w}_{k'}} \to |\Phi|^2 e^{-2\tilde{w}}$$

uniformly on $K$. Therefore, passing to the limit in

$$\Delta\tilde{w}_{k'} = e^{2\tilde{w}_{k'}} - |\Phi|^2 e^{-2\tilde{w}_{k'}},$$

we obtain

$$\Delta\tilde{w} = e^{2\tilde{w}} - |\Phi|^2 e^{-2\tilde{w}}$$

on $K$. Since $K$ is arbitrary, the equation holds on all of $\mathbb{D}$.

Finally, passing to the limit in (8.1), we get

$$e^{2\tilde{w}} \geq \frac{4}{(1-|z|^2)^2}.$$

Hence the metric $e^{2\tilde{w}}|dz|^2$ dominates the Poincaré metric on $\mathbb{D}$. Therefore every curve which has infinite length with respect to the Poincaré metric also has infinite length with respect to $e^{2\tilde{w}}|dz|^2$. Since the Poincaré metric is complete, this proves that $e^{2\tilde{w}}|dz|^2$ is complete.

This concludes the proof. □

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